Elastic Net

Elastic Net Regression will teach us how to build predictive models when many potential predictors are correlated and some may contain redundant information. It combines the L1 penalty used by LASSO with the L2 penalty used by Ridge Regression, allowing coefficients to be both shrunk and reduced exactly to zero. Elastic Net balances feature selection with the more stable treatment of correlated predictors, helping us construct models that are flexible, interpretable and less prone to overfitting.

Problem	Problem title	Research-paper anchor	Dataset needed
Problem 14	Forecasting U.S. Industrial Production Growth	Stock and Watson's macroeconomic forecasting literature; Elastic Net methodology from Zou and Hastie	FRED monthly U.S. macroeconomic data
Problem 15	Forecasting U.S. Corporate Credit Spreads	Gilchrist and Zakrajšek's credit-spread/business-cycle literature; Elastic Net methodology from Zou and Hastie	FRED corporate bond yields, Treasury yields, policy rates, labour-market and financial-risk indicators
Problem 16	Forecasting the Renewable-Energy Transition Across Countries	Directed technical-change / environmental-innovation literature, especially Acemoglu et al.; Elastic Net methodology from Zou and Hastie	Our World in Data Energy + CO_2 datasets

Table 1: *Overview of Problems, Research Anchors, and Datasets*

14. Forecasting U.S. Industrial Production Growth

Problem 14: Forecasting U.S. Industrial Production Growth Using Correlated Macroeconomic Indicators

Research Anchor 1: The macroeconomic forecasting motivation comes from the work of Stock and Watson on forecasting economic activity using large sets of economic indicators.

Research Anchor 2: Zou and Hastie (2005), "Regularization and Variable Selection via the Elastic Net".

Dataset: FRED monthly U.S. macroeconomic data.

Can Elastic Net improve forecasts of future U.S. industrial-production growth by handling groups of correlated macroeconomic predictors more effectively than OLS, Ridge and LASSO?

The important feature is that many predictors will be **correlated**.

We will compare: *OLS vs Ridge vs LASSO vs Elastic Net.*

Suppose two variables are highly correlated: $X_1 \approx X_2$

LASSO may retain and eliminate, even if both represent economically related information.

Ridge handles correlated variables more smoothly, but usually does not perform feature selection.

Elastic Net combines both approaches.

Dependent Variable: $Future\ IP\ Growth_t = 100\ [ln(IP_{t+12}) - ln(IP_t)]$

We will use 12 predictors:

1. Current industrial-production growth
2. Unemployment
3. Payroll growth
4. Capacity utilisation
5. Inflation
6. Money-supply growth
7. Housing-start growth
8. Federal Funds Rate

9. 2-year Treasury yield
10. 10-year Treasury yield
11. Term spread
12. Corporate default spread

Elastic Net:

$$\min_{\beta} \left[RSS + \lambda \left(\rho \sum |\beta_j| + (1-\rho) \sum \beta_j^2 \right) \right]$$

Approximately: *l1_ratio=1*, means pure LASSO.

Approximately *l1_ratio=0*, means Ridge-like regularisation.

Results and Interpretation

The final forecasting exercise uses **460 training observations from June 1976 to September 2014** and **118 out-of-sample observations from October 2015 to July 2025**. A twelve-month gap separates the training and test samples because the dependent variable measures industrial-production growth over the subsequent twelve months. The average industrial-production growth rate in the training sample is **2.1020%**, which provides the historical-mean forecasting benchmark.

The predictor correlation matrix confirms substantial multicollinearity among the macroeconomic variables. In particular, the Federal Funds Rate and the two- and ten-year Treasury yields form a strongly correlated interest-rate group, while industrial-production growth, payroll growth and capacity utilisation also contain overlapping information about the business cycle. This makes the problem a useful setting for comparing how OLS, Ridge, LASSO and Elastic Net handle correlated predictors.

Cross-validation selects a Ridge alpha of **2898.58**, a LASSO alpha of **1.5039**, and an Elastic Net alpha of **1.5039** with an optimal $l1_ratio$of **0.90**. The original Ridge search returned the upper boundary of the initial alpha grid, so I expanded the search range before finalising the model. The revised Ridge optimum of 2898.58 lies within the

expanded range and is used in the final results. The large Ridge alpha indicates that cross-validation favours substantial coefficient shrinkage, while the Elastic Net mixing parameter is close to one, meaning that the selected Elastic Net specification is predominantly LASSO-like and places relatively little weight on its Ridge component.

Both LASSO and Elastic Net produce extremely sparse models. From twelve candidate macroeconomic predictors, each retains only **housing-start growth** and eliminates the remaining eleven variables. The standardised housing-growth coefficient equals **0.4604 under LASSO** and **0.5309 under Elastic Net**. I interpret this as evidence that housing activity is the most robust individual forecasting variable under the strong L1-dominated regularisation selected through time-series cross-validation. However, this coefficient should not be interpreted causally; it represents a predictive association rather than an estimate of the structural effect of housing activity on future industrial production.

The coefficient comparison illustrates how the different methods respond to multicollinearity. OLS assigns sizeable coefficients to several related interest-rate variables, including -4.6094 for the Federal Funds Rate, 2.0540 for the two-year Treasury yield and 1.8909 for the ten-year Treasury yield. Ridge retains these correlated predictors but substantially shrinks their coefficient magnitudes, reflecting the tendency of the L2 penalty to distribute predictive information across related variables rather than eliminating individual members of the group. By contrast, LASSO and Elastic Net eliminate the entire interest-rate group. This difference is central to the exercise: Ridge preserves correlated information, whereas the L1-dominated models impose much stronger sparsity.

The forecasting results show that greater sparsity does not improve performance in this application. **OLS is the strongest overall forecasting model**, achieving the lowest RMSE of **3.6933**, the lowest MAE of **2.4994**, and the highest conventional test-sample R^2 of **0.1733**. Ridge performs less strongly, with an RMSE of **4.3290**, an MAE of **2.6662**, and an R^2 of **-0.1358**. Nevertheless, Ridge still improves upon the historical-mean

benchmark, which records an RMSE of **4.4483** and an MAE of **2.8342**. These results suggest that substantial shrinkage provides some benefit relative to a simple constant forecast, but unrestricted OLS uses the available joint predictor information more effectively in the final test period.

LASSO and Elastic Net both perform substantially worse. LASSO produces an RMSE of **4.5045**, an MAE of **2.8532**, and an $R^2$of **-0.2298**, while Elastic Net records an RMSE of **4.5147**, an MAE of **2.8591**, and an $R^2$of **-0.2353**. Both sparse specifications underperform even the historical-average benchmark according to the principal forecasting metrics.

The benchmark-relative out-of-sample R^2values provide particularly clear evidence. OLS achieves an $R^2_{OOS}$of **0.3106**, indicating a substantial reduction in squared forecast errors relative to the historical-mean benchmark. Ridge also produces a positive R^2_{OOS}, although at the considerably smaller value of **0.0529**. Ridge reduces squared forecast error by approximately 5.3% relative to the historical-mean benchmark, whereas OLS achieves a much larger improvement. By contrast, LASSO records an $R^2_{OOS}$of **-0.0255** and Elastic Net records **-0.0301**, indicating that both sparse models perform worse than the historical-average forecast.

Elastic Net also fails to improve upon pure LASSO. Its RMSE is approximately **0.23% worse** than the LASSO RMSE. This result is consistent with the selected $l1_ratio$of 0.90. Because the Elastic Net penalty is heavily weighted toward its L1 component, its final behaviour closely resembles LASSO. Both models select the same single predictor and consequently generate very similar forecasts.

The coefficient-path figure confirms this interpretation. At very small penalty values, many predictors receive non-zero coefficients. As alpha increases, most coefficients progressively shrink toward zero, while housing growth remains active for longer than the other predictors. At the selected Elastic Net alpha of **1.5039**, virtually the entire predictor set has been eliminated. The cross-validation curves also show that

the selected solution lies in a region of relatively strong regularisation and that the optimal mixing parameter is strongly LASSO-like.

The actual-versus-forecast plot reveals the principal challenge facing all models. Industrial-production growth undergoes an extraordinary collapse associated with the COVID-19 pandemic, followed by a rapid rebound. Because predictor observations around 2019 are used to forecast growth over the subsequent twelve months, actual future industrial-production growth falls to approximately **-19%**, while none of the models anticipates the full contraction. The subsequent rebound produces the opposite problem, with realised growth exceeding **15%** while all models substantially underpredict the recovery.

These structural breaks are also visible in the Elastic Net residuals. Forecast errors become extremely negative during the production collapse, indicating severe overprediction, and then strongly positive during the rebound, indicating severe underprediction. Residuals subsequently become smaller but remain somewhat clustered rather than behaving entirely randomly around zero, suggesting that the forecasting relationship is not stable across all economic regimes.

The cumulative squared-error comparison further illustrates the difference between the regularisation methods. Ridge ultimately produces a positive forecasting gain relative to the historical-mean benchmark, which is consistent with its positive out-of-sample R^2. However, this gain is modest relative to OLS. LASSO and Elastic Net remain below zero, confirming that their extensive feature elimination generates cumulative forecasting losses relative to the benchmark.

(Refer to Appendix A.14 for more on Results)

Conclusion

Overall, I find that the strongly correlated macroeconomic predictors contain useful joint forecasting information that is lost when L1 regularisation becomes too aggressive. **OLS provides the strongest overall out-of-sample performance**, while

Ridge retains the complete predictor set, applies substantial shrinkage and still improves upon the historical-average benchmark. In contrast, LASSO and Elastic Net reduce the model to housing-start growth alone and both underperform the historical mean.

The exercise demonstrates an important feature of Elastic Net: combining L1 and L2 penalties does not guarantee superior forecasting performance. The usefulness of Elastic Net depends on the particular combination of alpha and l1_ratioselected from the data. In this application, the selected l1_ratioof **0.90** causes Elastic Net to behave very similarly to LASSO, so it inherits LASSO's aggressive sparsity rather than exploiting the group-preserving advantage of Ridge. The results also show that a more advanced regularisation method does not necessarily outperform a simpler unrestricted model when the predictor relationships remain useful out of sample.

15. Forecasting U.S. Corporate Credit Spreads

Problem 15: Forecasting U.S. Corporate Credit Spreads Using Macroeconomic and Financial Risk Indicators

Research Anchor: Linking corporate bond spreads and credit conditions to macroeconomic activity, especially **Gilchrist and Zakrajšek**.

Which macroeconomic and financial indicators help forecast changes in U.S. corporate credit spreads over the next six months, and does Elastic Net outperform Ridge and LASSO when predictors contain substantial overlapping information?

Can Elastic Net forecast whether U.S. corporate credit conditions will tighten or deteriorate over the next six months when many interest-rate, credit-risk and macroeconomic predictors are strongly correlated?

When investors become more concerned about credit risk, yields on lower-quality corporate bonds typically rise relative to safer bonds.

A simple measure is the spread between Moody's BAA and AAA corporate bond yields.

We will use 13 predictors.

Credit-market information

- Current default spread
- 3-month default-spread change, 12-month default-spread change
- AAA corporate yield, BAA corporate yield

Interest-rate information

- Federal Funds Rate, Treasury term spread
- 2-year Treasury yield, 10-year Treasury yield

Macroeconomic information

- Unemployment rate, Industrial-production growth, Inflation

Financial-risk information

- VIX

Results and Interpretation

The corporate credit-spread forecasting exercise examines whether macroeconomic and financial indicators contain useful information for forecasting changes in the U.S. corporate default spread over the subsequent six months. The **dependent variable** is defined as the six-month-ahead change in the difference between BAA and AAA corporate bond yields.

Positive values represent future spread widening and deterioration in credit conditions, while negative values represent future spread narrowing.

The target series demonstrates that most six-month changes in credit spreads are relatively small but are interrupted by periods of substantial financial stress. The most pronounced movements occur around the **Global Financial Crisis**, when future spread changes approach approximately two percentage points in both directions, while smaller but still meaningful fluctuations occur around subsequent periods of market stress.

This pattern suggests that changes in credit conditions are difficult to forecast because relatively tranquil periods are occasionally interrupted by large and abrupt movements.

The historical mean future spread change in the training sample is only **-0.0033 percentage points**, which is economically close to zero. Consequently, the historical-mean benchmark effectively predicts very little change in corporate spreads over the subsequent six months. This creates a demanding benchmark because any predictor-

based model must consistently forecast changes in spreads more accurately than simply assuming that credit conditions will remain approximately unchanged.

The correlation analysis confirms **extremely strong multicollinearity** among the financial predictors. The correlation between AAA and BAA corporate yields is approximately **0.975**, while BAA yields and the ten-year Treasury yield have a correlation of approximately **0.972**. The Federal Funds Rate and two-year Treasury yield are correlated at approximately **0.956**, while the two- and ten-year Treasury yields have a correlation of approximately **0.904**.

These relationships indicate that several variables contain closely overlapping information about the level of interest rates and financial conditions. The dataset provides a particularly relevant setting for comparing OLS with regularisation methods designed to handle correlated predictors.

The regularisation results reveal a strong preference for shrinkage. LASSO selects an optimal alpha of **0.2154**, while Elastic Net selects an alpha of **0.2477** and an $l1_ratio$of **0.90**. The Elastic Net specification is predominantly LASSO-like. Both models eliminate **all thirteen candidate predictors**, reducing the final forecasting model to an intercept-only specification.

Ridge displays an even more extreme form of shrinkage. After the initial optimum occurred at the boundary of the search range, I expanded the Ridge alpha grid substantially. Cross-validation then selected the new upper boundary of **1,000,000**. At this penalty strength, all Ridge coefficients become economically negligible, with most rounded to 0.0000. The largest remaining coefficients are only approximately -0.0001 for the current default spread and 0.0001 for inflation. The model has effectively converged toward the historical-mean forecast. Rather than continuing to expand the alpha grid indefinitely, I interpret this boundary result as evidence that cross-

validation favours the limiting Ridge solution in which predictor coefficients approach zero.

This result contrasts sharply with OLS. OLS assigns non-trivial coefficients to numerous variables, including **-0.2912 for the current default spread, 0.1808 for inflation, -0.1260 for the BAA yield, 0.1007 for the term spread and 0.0970 for the VIX**. Given the extreme correlation among several interest-rate and credit variables, these unrestricted coefficients appear unstable out of sample. Ridge, LASSO and Elastic Net all respond by imposing very strong shrinkage, although Ridge approaches zero continuously while LASSO and Elastic Net set the coefficients exactly to zero.

The forecast-performance results strongly favour simple, heavily regularised specifications over OLS. The historical-mean benchmark records an RMSE of **0.2350**, an MAE of **0.1760**, and a conventional test-sample R^2 of **-0.0240**.

LASSO and Elastic Net generate exactly the same values because both reduce to intercept-only models and reproduce the historical-mean forecast.

Ridge performs almost identically, with an RMSE of **0.2350**, an MAE of **0.1760**, and an R^2 of **-0.0232**. Its benchmark-relative out-of-sample R^2 is only **0.0008**, implying an improvement in squared forecasting error of approximately **0.08%** relative to the historical mean. Although technically positive, this improvement is economically negligible. I regard Ridge as effectively equivalent to the historical-mean benchmark in this application.

OLS performs dramatically worse. Its RMSE rises to **0.5283** and its MAE to **0.4646**, while its conventional R^2 falls to **-4.1741**. Its benchmark-relative R^2_{OOS} equals **-4.0529**, indicating severe deterioration relative to the historical-average forecast. This result suggests that attempting to estimate unrestricted individual effects from the highly correlated predictor set leads to substantial overfitting and poor generalisation.

The actual-versus-forecast figure reinforces this conclusion. The realised six-month change in credit spreads displays meaningful positive and negative movements, including a sharp widening around the onset of the COVID-19 period and subsequent reversals. By contrast, the historical mean, LASSO and Elastic Net forecasts remain essentially flat around zero. Ridge also becomes almost indistinguishable from these constant forecasts after the expanded alpha search. The models conclude that the predictable component of future spread changes is extremely small relative to realised variation.

The conventional direction-accuracy results require particularly careful interpretation. The historical mean, Ridge, LASSO and Elastic Net each achieve an accuracy of **67.86%,** while OLS achieves only **35.71%**. However, this apparently strong 67.86% result does not indicate sophisticated market timing. Because the historical mean is slightly negative, the benchmark effectively predicts spread narrowing in every test observation. Ridge, LASSO and Elastic Net behave similarly following their extreme shrinkage. Their direction accuracy largely reflects the fact that approximately two-thirds of test-period observations experienced spread narrowing.

The widening-recall results expose the limitation of these apparently high direction-accuracy figures. Ridge, LASSO and Elastic Net all record a widening recall of **0%**. In other words, none of the heavily regularised models identifies any of the occasions when credit spreads subsequently widen. From a credit-risk perspective, this is a significant limitation because spread-widening episodes represent precisely the periods of deteriorating financial conditions that may be most important to anticipate.

OLS produces the opposite pattern. It achieves **100% widening recall**, meaning that every actual widening episode is identified, but its overall direction accuracy is only **35.71%** and its forecast-error measures are extremely poor. This suggests that OLS predicts widening too frequently and generates many false alarms. The result

illustrates why recall cannot be interpreted independently of overall forecasting performance: **detecting every stress event is of limited value if the model repeatedly predicts stress when it does not subsequently occur.**

The Elastic Net coefficient plot and the LASSO-versus-Elastic Net comparison visually confirm that all coefficients are exactly zero. The final models contain no predictor-specific information beyond the intercept. The coefficient-path figure shows that numerous variables receive non-zero coefficients at very weak penalty values, but these progressively shrink as alpha increases. By the cross-validation-selected Elastic Net alpha of **0.2477**, all variables have been eliminated.

The Elastic Net cross-validation curves provide further evidence for strong regularisation. Cross-validated RMSE falls substantially as alpha increases from very small values, showing that weakly regularised models generalise poorly. The error curves then flatten near their minimum across several values of $l1_ratio$. Although the selected $l1_ratio$ is 0.90, the differences between several penalty mixtures are relatively small near the optimum. I interpret the main finding as a preference for strong regularisation rather than as evidence that the precise 90% L1 weighting is uniquely optimal.

The forecast-error plot also reflects the intercept-only nature of Elastic Net. Because the model predicts almost no future spread change, its residuals closely track realised movements in the target variable. Large positive errors occur when spreads unexpectedly widen, while large negative errors occur when spreads narrow more strongly than expected. The **largest errors arise during periods of abrupt changes in financial conditions**, precisely when a credit-risk model would ideally provide the greatest forecasting value.

The cumulative forecast-performance comparison ultimately reinforces the same conclusion. Once Ridge is estimated with the expanded alpha grid, its forecasting

performance becomes almost indistinguishable from the historical mean, consistent with its $R^2_{OOS}$of only **0.0008**. LASSO and Elastic Net coincide exactly with the benchmark. The predictor-based regularised models provide essentially no economically meaningful forecasting improvement over assuming that future credit-spread changes will be approximately zero.

(Refer to Appendix A.15 for more on Results)

Conclusion

Overall, I find little evidence that the thirteen macroeconomic and financial predictors provide stable incremental information for forecasting six-month changes in U.S. corporate credit spreads within this specification.

OLS attempts to exploit the available predictors but performs extremely poorly out of sample, which is consistent with the substantial multicollinearity among interest-rate and corporate-yield variables. Regularisation prevents this severe overfitting, but cross-validation ultimately favours penalties so strong that Ridge becomes almost an intercept-only model while LASSO and Elastic Net eliminate every predictor completely.

The results illustrate an important role of regularisation that goes beyond improving forecasts. **Regularisation can reveal when a rich predictor set contains insufficient stable information to justify a complex model.** In this case, the appropriate response to multicollinearity and unstable forecasting relationships is not to retain a carefully selected subset of variables, but effectively to discard their incremental contribution altogether.

At the same time, the direction and widening-recall results demonstrate an important practical limitation. The highly regularised models achieve reasonable overall

direction accuracy only because spread narrowing occurs more frequently in the test sample. They fail to identify any future widening episodes.

For credit-risk applications, forecast evaluation should consider not only average statistical accuracy but also performance during periods of deteriorating financial conditions, since these events may carry substantially greater economic importance.

16. Forecasting the Renewable-Energy Transition Across Countries

Problem 16: Forecasting the Renewable-Energy Transition Across Countries Using Economic, Emissions and Energy-Mix Indicators

Research Anchor: Acemoglu, Aghion, Bursztyn and Hemous (2012), who model technological change as being directed toward either "clean" or "dirty" technologies and show how policy and innovation incentives can influence the transition between them.

Dataset: Our World in Data Energy dataset. The dataset combines sources including Ember, the Energy Institute and the U.S. Energy Information Administration.

Which existing characteristics of a country's electricity and energy system help predict how quickly the share of renewable electricity will increase in the following year?

Can Elastic Net forecast the next-year change in a country's renewable-electricity share using highly correlated economic, energy-mix and carbon-intensity indicators?

Does Elastic Net handle the correlated electricity-mix variables more effectively than LASSO or Ridge?

Let:

$$R_{i,t} = \textit{renewable share of electricity generation}$$

for country iin year t.

Our target is:

$$\boxed{\Delta R_{i,t+1} = R_{i,t+1} - R_{i,t}}$$

measured in **percentage points**.

If $\Delta R > 0$, the country's renewable share increased.

If $\Delta R < 0$, the country's renewable share decreased.

So, we are modelling the **speed and direction of the electricity transition**, not simply the current renewable level.

We will use 16 predictors.

Existing electricity mix

- renewables_share_elec , renewable_change , fossil_share_elec
- coal_share_elec , gas_share_elec
- solar_share_elec , wind_share_elec , nuclear_share_elec

Electricity and energy-system characteristics

- carbon_intensity_elec
- electricity_demand_growth , electricity_per_capita
- energy_per_capita , energy_per_gdp

Economic and demographic characteristics

- log_gdp_per_capita , population_growth , electricity_growth

Several of these are very closely related by construction.

Results and Interpretation

The renewable-energy transition exercise examines whether electricity-mix, energy-system, carbon-intensity, economic and demographic indicators can help predict the next-year change in a country's renewable share of electricity generation. A positive value of the dependent variable represents an increase in the renewable-electricity share, while a negative value represents a decline.

The historical mean annual renewable-share change in the training sample is **0.1612 percentage points**, which provides the benchmark forecast against which I compare the alternative models.

The distribution of annual renewable-share changes is strongly concentrated around zero but displays long tails in both directions. Most country-year observations experience relatively modest annual changes in their electricity mix, while a much smaller number undergo very large increases or decreases.

The presence of these extreme observations suggests that **gradual changes in renewable shares may be more predictable than abrupt transformations in national electricity systems.**

The evolution of the average renewable transition through time also shows that the pace of change is not constant. Average next-year renewable-share growth is relatively weak and occasionally negative during the earlier part of the sample, while it becomes predominantly positive from the mid-2010s onward. A particularly large average increase occurs around 2019. This indicates that the energy transition accelerated over parts of the sample and that forecasting the process requires more than simply assuming a constant long-run rate of renewable adoption.

The predictor correlation matrix confirms **substantial multicollinearity among the energy-transition variables.** Renewable-electricity share and fossil-electricity share have a correlation of approximately **-0.931**, while fossil-electricity share and electricity carbon intensity are correlated at approximately **0.929**. Renewable-electricity share and carbon intensity are also strongly negatively correlated at approximately **-0.868**. In addition, electricity consumption per capita and energy consumption per capita have a correlation of approximately **0.833**, while energy consumption per capita and log GDP per capita have a correlation of approximately **0.706**. These relationships indicate that many predictors contain overlapping information about the same underlying electricity and energy system.

The consequences of this multicollinearity are particularly visible in the OLS coefficient estimates. OLS produces extremely large standardised coefficients of approximately **5759.85 for renewable-electricity share**, **5846.27 for fossil-electricity**

share, and **2194.14 for nuclear-electricity share**. I do not interpret these coefficients economically because the electricity-share variables are closely related by construction. The extreme magnitudes instead demonstrate how unstable unrestricted OLS coefficients can become when several predictors contain nearly the same information. This provides a strong rationale for applying regularisation.

Cross-validation selects an optimal Ridge alpha of **1686.13** and an optimal LASSO alpha of **0.1417**. For Elastic Net, the optimal alpha is **0.7565**, with an $l1_ratio$ of **0.05**. The selected $l1_ratio$ is particularly important because it places the Elastic Net specification very close to the Ridge end of the Ridge-LASSO spectrum.

In contrast to the more LASSO-like Elastic Net specifications obtained in some of the preceding exercises, the present model strongly favours the L2 component of the penalty. This suggests that preserving groups of correlated energy-system variables is useful in this application.

The difference between LASSO and Elastic Net feature selection illustrates this point clearly. LASSO retains only 6 of the 16 candidate predictors. Elastic Net, by contrast, retains **15 of the 16 predictors** and eliminates only **coal-electricity share**.

I observe a clear grouping effect. Rather than selecting a small number of representatives from highly correlated groups, Elastic Net keeps most related variables in the model while shrinking their individual coefficients.

The largest positive Elastic Net coefficient is associated with **wind-electricity share**, at **0.2448**, while solar-electricity share receives a coefficient of **0.1297**. These results indicate that countries with larger existing wind and solar components tend, conditional on the other variables, to experience somewhat larger subsequent increases in their overall renewable-electricity share. I interpret these as predictive relationships rather than causal effects.

The strongest negative Elastic Net coefficient is the previous year's renewable-share change, at **-0.2867**. This suggests some degree of mean reversion in the speed of the

renewable transition. A country experiencing an unusually large increase in renewable share in one year is less likely to experience an equally large increase immediately afterward.

Current renewable-electricity share also receives a negative coefficient of **-0.1548**, which is consistent with the possibility that countries already operating with relatively high renewable shares have less scope for very large additional annual increases.

Electricity-demand growth and electricity growth receive coefficients of **-0.1604** and **-0.1419**, respectively. One possible interpretation is that rapidly expanding electricity demand may be more difficult to satisfy entirely through new renewable capacity, although the model does not establish the causal mechanism behind this association.

Fossil-electricity share receives a positive coefficient of **0.1154**, while carbon intensity receives a positive coefficient of **0.0557**. These apparently counterintuitive signs require particular caution because fossil share, renewable share and carbon intensity are extremely highly correlated.

I interpret them only as conditional predictive coefficients after controlling for the rest of the energy system, rather than as evidence that fossil dependence or carbon intensity causes a faster renewable transition.

The LASSO-versus-Elastic-Net coefficient comparison reinforces the importance of the grouping effect. LASSO eliminates fossil share, gas share, nuclear share, carbon intensity, electricity consumption per capita, energy consumption per capita, energy intensity, GDP per capita and population growth. Elastic Net retains all of these variables except coal share, although many receive relatively small coefficients.

This is consistent with the theoretical motivation for Elastic Net: when predictors contain related information, its Ridge component allows the model to distribute predictive weight across the group rather than forcing the selection of only a few variables.

The coefficient-path figure tells a similar story. At very small penalty values, most variables receive non-zero coefficients. As alpha increases, the coefficient magnitudes gradually shrink toward zero. At the cross-validation-selected alpha, however, most variables remain active because the optimal $l1_ratio$is only 0.05. The model achieves regularisation primarily through coefficient shrinkage rather than extensive variable elimination.

The cross-validation-error figure provides additional evidence in favour of the strongly Ridge-like specification. The lowest cross-validated RMSE occurs for the $l1_ratio = 0.05$curve around the selected alpha of 0.7565. The minima for larger L1 ratios are very close, but the lowest point is achieved when the Ridge component dominates. I interpret the result as evidence that preserving correlated groups of energy-transition indicators provides a small advantage within the training sample.

Out-of-sample forecasting performance improves modestly under regularisation. The historical-mean benchmark produces an RMSE of **4.7964**, an MAE of **2.6999**, and an R^2 of **-0.0255**. OLS improves RMSE slightly to **4.7517**, but its MAE increases to **2.7388**, and its R^2remains slightly negative at **-0.0065**. Ridge performs more strongly, with an RMSE of **4.6890**, an MAE of **2.6481**, and an $R^2$of **0.0199**.

LASSO achieves the strongest aggregate forecasting performance, with an RMSE of **4.6826**, an MAE of **2.6329**, and an R^2 of **0.0226**. Elastic Net follows almost identically, with an RMSE of **4.6843**, an MAE of **2.6344**, and an $R^2$of **0.0219**. The difference in RMSE between LASSO and Elastic Net is only **0.0017 percentage points**, equivalent to approximately **0.04%**. I find no economically meaningful difference in aggregate forecast accuracy between the two models, despite their substantially different feature-selection behaviour.

The benchmark-relative out-of-sample R^2results tell the same story. OLS produces an $R^2_{OOS}$of **0.0186**, Ridge achieves **0.0443**, LASSO achieves **0.0469**, and Elastic Net achieves **0.0462**. LASSO reduces squared forecast error by approximately 4.69% relative to the

historical-mean benchmark, while Elastic Net reduces it by approximately 4.62%. The gains are modest, but the positive values show that the regularised models contain useful predictive information beyond simply forecasting the long-run average renewable-share change.

The yearly-average forecast comparison shows that the regularised models capture the broader acceleration in renewable deployment considerably better than the fixed historical-mean benchmark. The historical mean remains at approximately **0.161 percentage points**, whereas realised average renewable-share growth during the test period is generally much larger. Ridge, LASSO and Elastic Net forecast positive increases of a more realistic magnitude and track the broad direction of the transition more effectively.

The largest forecasting miss occurs around **2019**, when the realised average next-year renewable-share increase rises to approximately **1.74 percentage points**. The regularised models forecast only around 0.65 to 0.70 percentage points. They correctly anticipate a positive renewable transition but substantially underestimate its magnitude. Forecasts are considerably closer to realised average changes in 2020 and 2021, while by 2022 the regularised models tend to overpredict the average increase.

Direction accuracy differs very little across models. The historical mean achieves **56.02%**, OLS **56.27%**, Ridge **56.27%**, LASSO **56.40%**, and Elastic Net **56.27%**. LASSO records the highest value, although its advantage is negligible. The historical-mean result also requires caution because the benchmark is positive and essentially predicts that renewable share will increase in every observation. Its direction accuracy consequently reflects, to a substantial extent, the proportion of positive renewable transitions in the test sample rather than genuine variation in its forecasts.

The positive-transition recall measure provides somewhat stronger differentiation. OLS correctly identifies **83.14%** of observations in which renewable share subsequently rises, Ridge identifies **85.68%**, LASSO identifies **86.37%**, and Elastic Net

achieves the highest recall at **86.84%**. Elastic Net performs particularly well at identifying positive renewable-transition episodes, although recall alone does not capture how often the model incorrectly predicts a positive transition when renewable share subsequently falls.

The fossil-dependence analysis reveals a substantial difference in forecasting performance across electricity systems. Elastic Net records an RMSE of **5.3757** among observations where fossil fuels account for 70% or less of electricity generation, compared with an RMSE of only **3.6108** where fossil generation exceeds 70%. The model forecasts annual renewable-share changes considerably more accurately in the highly fossil-dependent group.

The corresponding forecast-error plot broadly supports this result. Errors are relatively widely dispersed across much of the intermediate fossil-share range, whereas many observations at very high fossil shares cluster closer to zero. Several large outliers remain, however, including some observations with large positive forecast errors.

I do not interpret the difference causally. Highly fossil-dependent countries may differ systematically in the volatility of their renewable transitions, income levels, electricity-market structures, policy environments and other characteristics. The result indicates a difference in forecastability rather than establishing that fossil dependence itself makes renewable transitions easier to predict.

The actual-versus-predicted scatter plot highlights one of the main limitations of Elastic Net. Actual renewable-share changes range from declines exceeding **20 percentage points** to increases above **30 percentage points**, while the model's predictions remain concentrated within a much narrower range around moderate values. If the forecasts were perfect, the observations would lie close to the 45-degree reference line. Instead, the predictions form a compressed cluster around the centre of the distribution.

This indicates substantial regression toward the mean. Elastic Net appears considerably better at forecasting ordinary annual changes than extreme transition episodes. Very large positive changes are substantially underpredicted, while extreme negative changes are forecast to be much closer to zero than they actually are. This helps explain why Elastic Net can outperform the historical-mean benchmark while still generating only a relatively small positive R^2.

The cumulative squared-error comparison confirms that Ridge, LASSO and Elastic Net all ultimately outperform the historical-mean benchmark. Their cumulative gains fluctuate considerably across individual country-year observations, meaning that the forecasting advantage is not uniform throughout the test sample. Nevertheless, all three regularised models finish substantially above zero. LASSO and Elastic Net remain particularly close, which is consistent with their almost identical overall RMSE values despite their very different coefficient structures.

(Refer to Appendix A.16 for more on Results)

Conclusion

Overall, I find that regularisation provides a modest but consistent improvement in forecasting next-year changes in renewable-electricity shares across countries. LASSO produces the strongest aggregate forecasting performance, while Elastic Net performs almost identically and Ridge follows closely behind.

The main advantage of Elastic Net in this application is not superior point-forecast accuracy, but its ability to retain groups of highly correlated energy-system variables while controlling their coefficient magnitudes.

The selected $l1_ratio$ of **0.05** provides particularly strong evidence of this behaviour. Elastic Net retains 15 of 16 predictors, compared with only six under LASSO, yet achieves virtually the same out-of-sample forecasting performance. If the objective is maximum parsimony together with slightly better RMSE, LASSO offers the more economical specification. If the objective also includes preserving groups of

economically related electricity, carbon and energy indicators, Elastic Net provides a more stable representation of the underlying system.

The exercise demonstrates both the strength and limitation of Elastic Net. It successfully addresses severe multicollinearity and avoids the unstable coefficient magnitudes produced by OLS, while preserving far more correlated information than LASSO.

However, this richer predictor structure does not automatically translate into better forecast accuracy. Extreme renewable-transition episodes also remain difficult to predict, suggesting that important policy, technological, institutional and country-specific shocks are not fully captured by the linear predictors included in the model.

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Acemoglu, D., Aghion, P., Bursztyn, L. and Hemous, D. (2012) 'The environment and directed technical change', *American Economic Review*, 102(1), pp. 131–166. doi: 10.1257/aer.102.1.131.

Zou, H. and Hastie, T. (2005) 'Regularization and variable selection via the elastic net', *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 67(2), pp. 301–320. doi: 10.1111/j.1467-9868.2005.00503.x.

Our World in Data (2026) *Energy dataset*. Our World in Data. Version 2026-04-24. Available at: Our World in Data Energy Dataset (Accessed: 30 August 2026). The dataset contains country-year information on electricity mix, renewable and fossil shares, energy use, carbon intensity, GDP and population used in this exercise.

Appendices

A.14 Forecasting U.S. Industrial Production Growth Using Correlated Macroeconomic Indicators

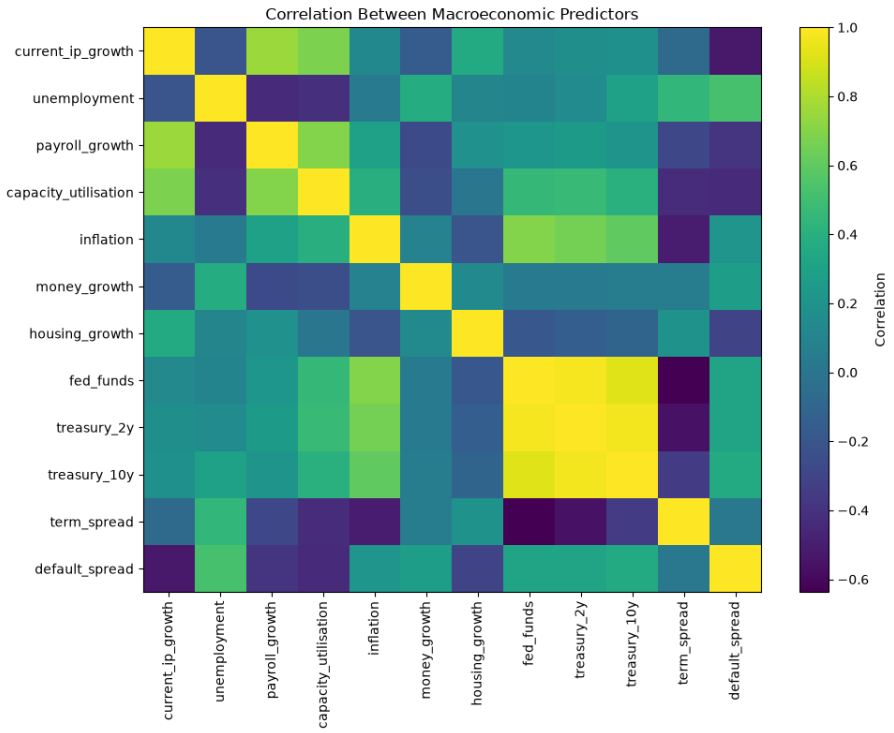


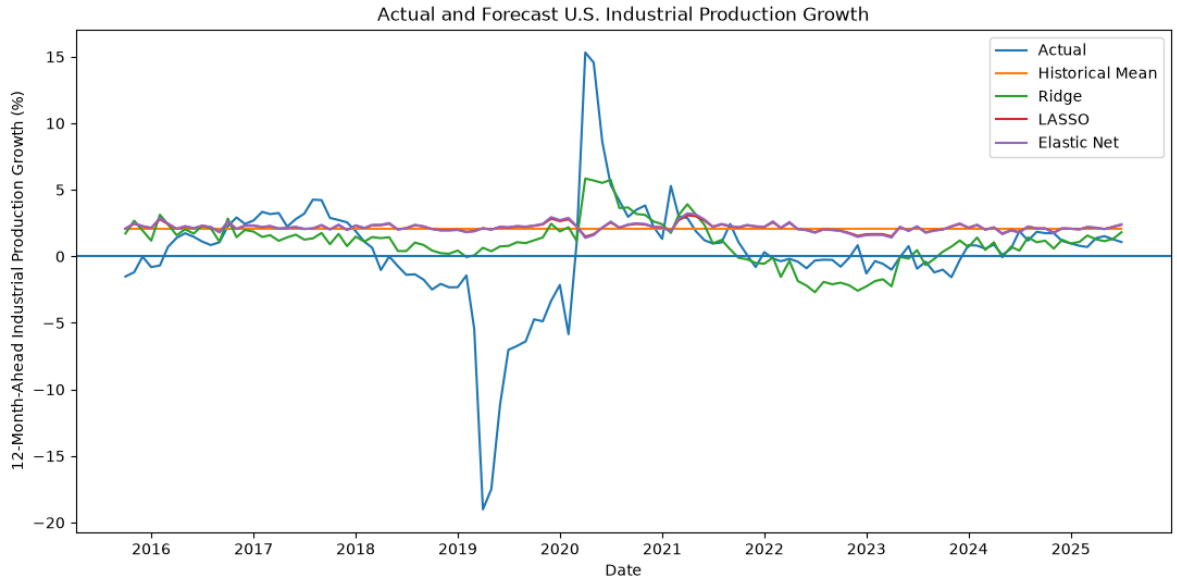
Figure 1: *Correlation Matrix*

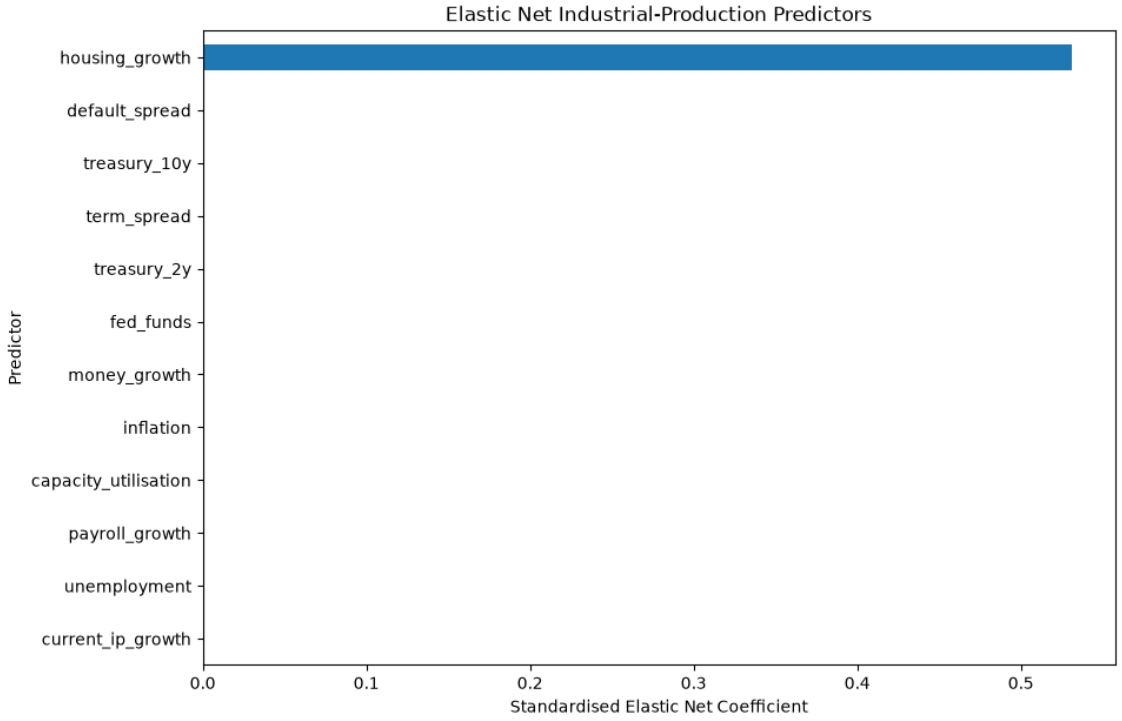
	OLS	Ridge	LASSO	Elastic_Net	LASSO_Selected	ElasticNet_Selected
current_ip_growth	-1.0266	-0.2718	0.0000	0.0000	False	False
unemployment	2.2771	1.1309	0.0000	0.0000	False	False
payroll_growth	0.0024	0.1825	0.0000	-0.0000	False	False
capacity_utilisation	0.9614	0.0855	0.0000	-0.0000	False	False
inflation	-1.5575	-1.1463	-0.0000	-0.0000	False	False
money_growth	0.2111	0.0153	0.0000	-0.0000	False	False
housing_growth	1.1957	1.1692	0.4604	0.5309	True	True
fed_funds	-4.6094	-0.4017	-0.0000	-0.0000	False	False
treasury_2y	2.0540	0.3938	0.0000	-0.0000	False	False
treasury_10y	1.8909	0.4303	0.0000	-0.0000	False	False
term_spread	-1.8538	-0.1369	0.0000	0.0000	False	False
default_spread	-1.2063	-0.6692	0.0000	-0.0000	False	False

Figure 2: *Coefficient Table for Industrial Production Growth*

Model	RMSE	MAE	R^2	R^2_{OOS}
Historical Mean	4.4483	2.8342	-0.1992	—
OLS	3.6933	2.4994	0.1733	0.3106
Ridge	4.3290	2.6662	-0.1358	0.0529
LASSO	4.5045	2.8532	-0.2298	-0.0255
Elastic Net	4.5147	2.8591	-0.2353	-0.0301

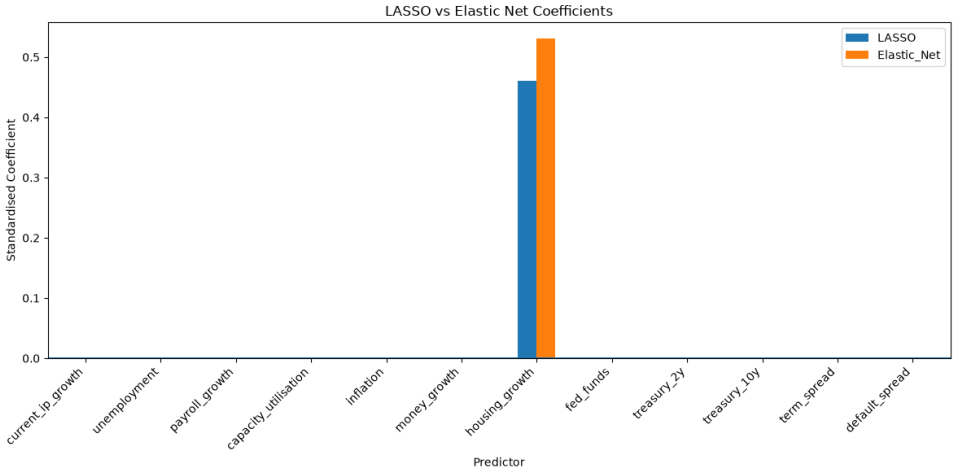
Table 2: *OOS Forecast Performance for US Industrial Production Growth*

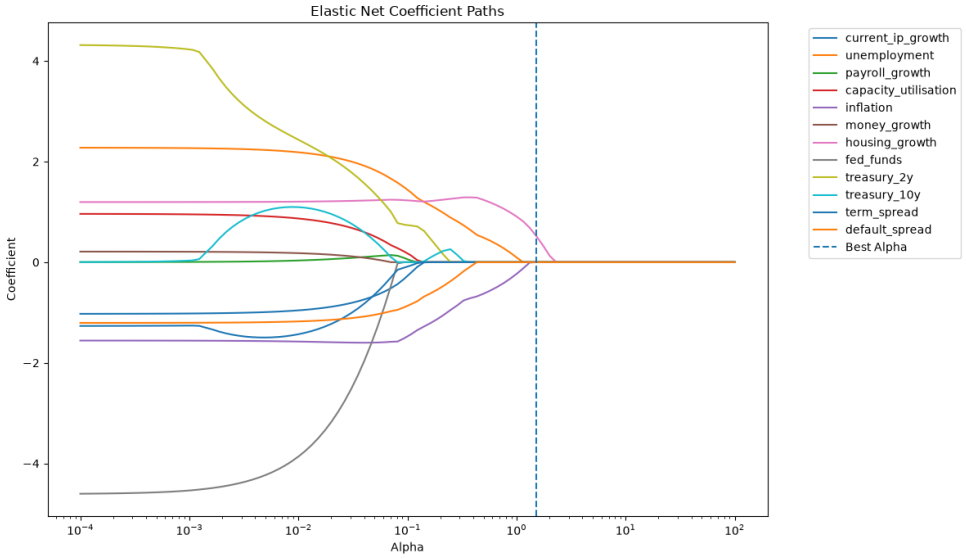


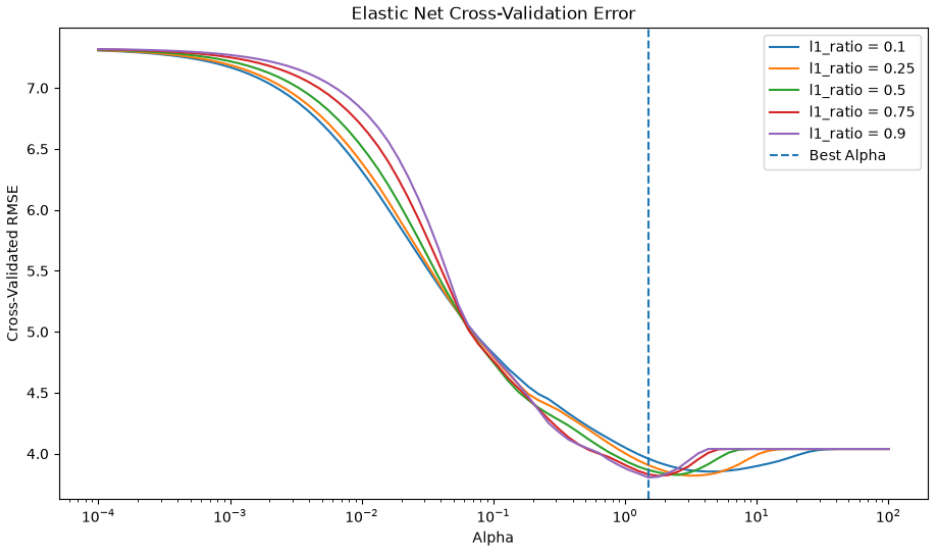


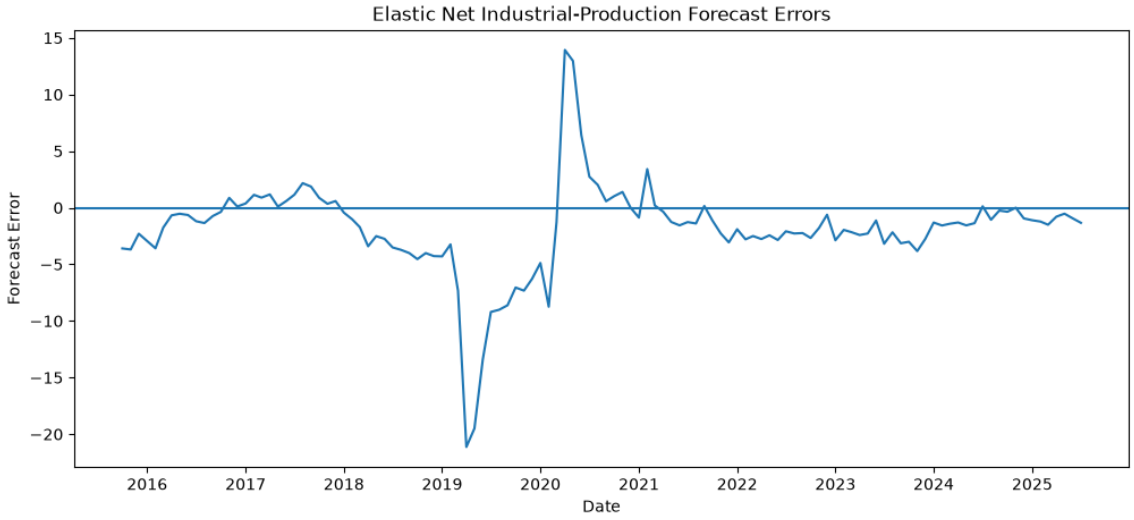
	LASSO	Elastic_Net
current_ip_growth	0.0000	0.0000
unemployment	0.0000	0.0000
payroll_growth	0.0000	-0.0000
capacity_utilisation	0.0000	-0.0000
inflation	-0.0000	-0.0000
money_growth	0.0000	-0.0000
housing_growth	0.4604	0.5309
fed_funds	-0.0000	-0.0000
treasury_2y	0.0000	-0.0000
treasury_10y	0.0000	-0.0000
term_spread	0.0000	0.0000
default_spread	0.0000	-0.0000

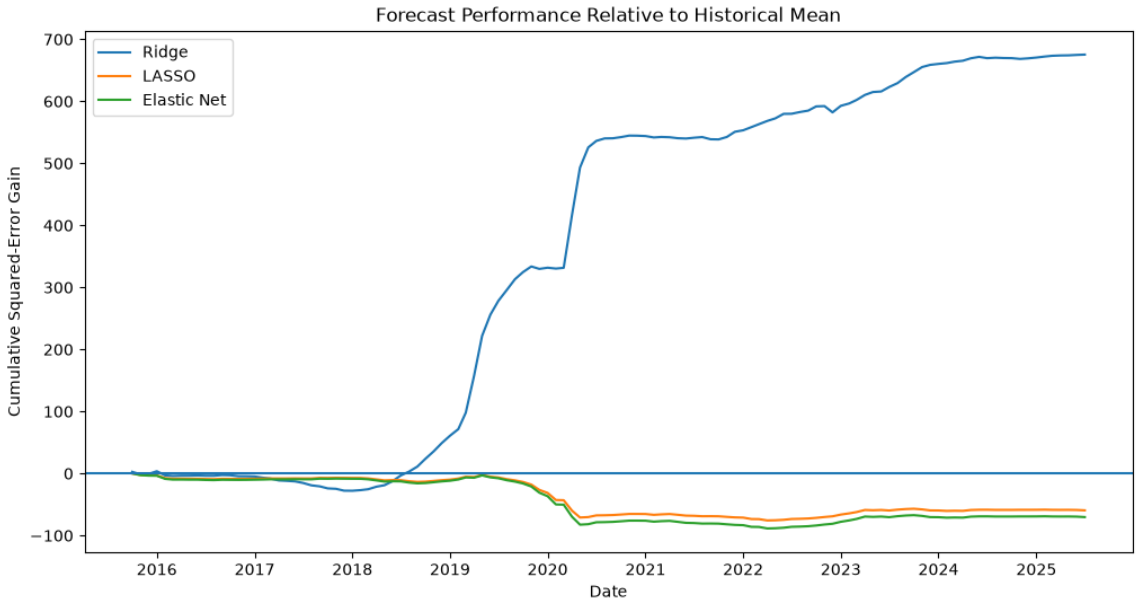
Figure 3: *LASSO versus Elastic Net Coefficients for US Industrial Production Growth*


LASSO vs Elastic Net Coefficients
LASSO
Elastic_Net
Standardised Coefficient
0.0
0.1
0.2
0.3
0.4
0.5
current_ip_growth
unemployment
payroll_growth
capacity_utilisation
inflation
money_growth
housing_growth
fed_funds
treasury_2y
treasury_10y
term_spread
default_spread
Predictor

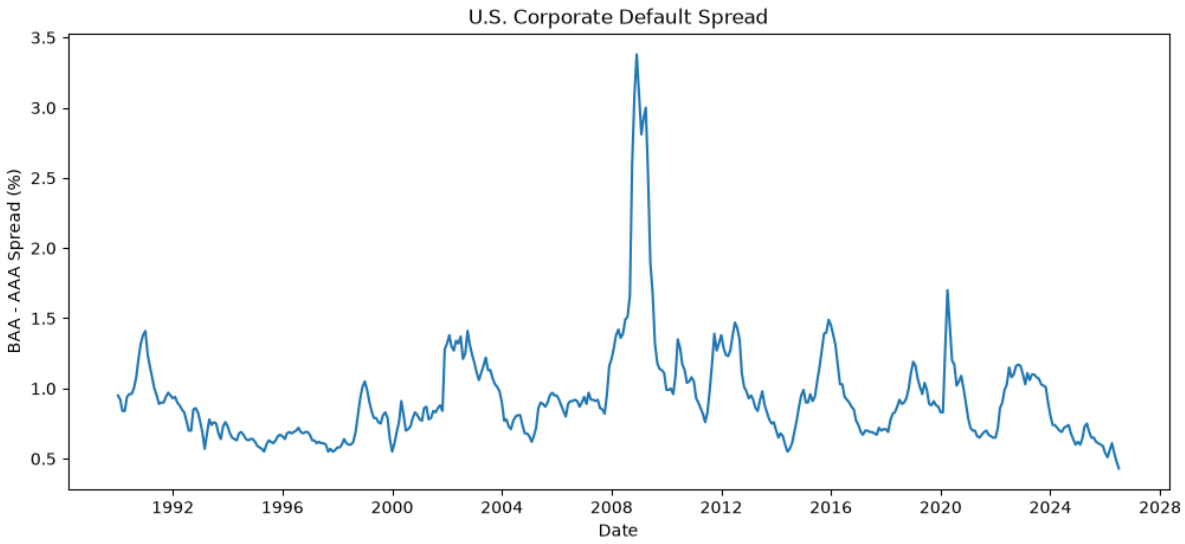

Elastic Net Coefficient Paths
Coefficient
Alpha
current_ip_growth
unemployment
payroll_growth
capacity_utilisation
inflation
money_growth
housing_growth
fed_funds
treasury_2y
treasury_10y
term_spread
default_spread
Best Alpha

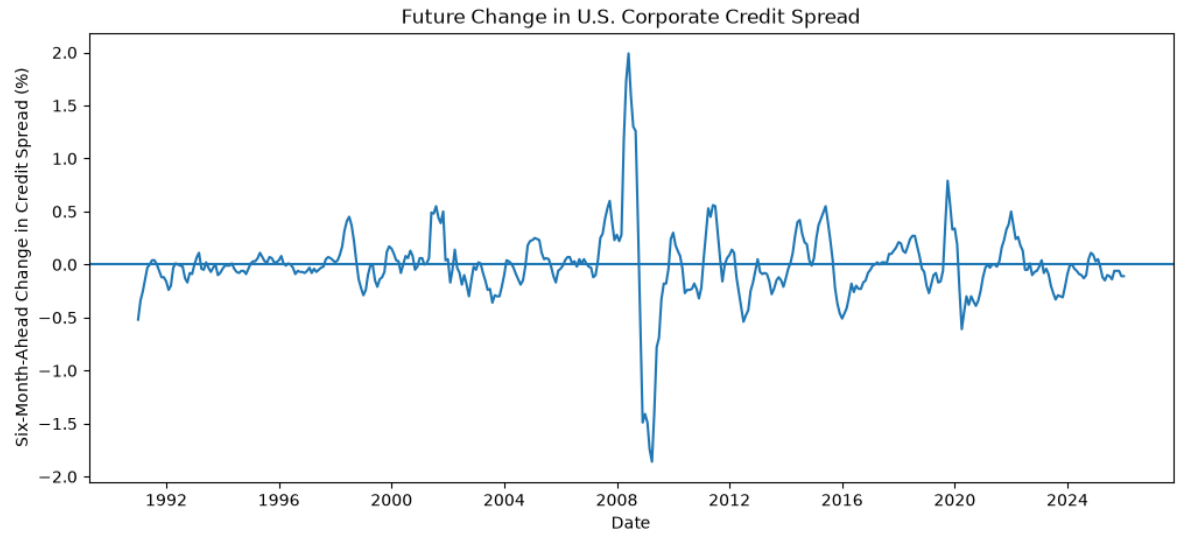

Elastic Net Cross-Validation Error
Cross-Validated RMSE
Alpha
l1_ratio = 0.1
l1_ratio = 0.25
l1_ratio = 0.5
l1_ratio = 0.75
l1_ratio = 0.9
Best Alpha


Elastic Net Industrial-Production Forecast Errors
Forecast Error
15
10
5
0
−5
−10
−15
−20
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
Date


Forecast Performance Relative to Historical Mean
Ridge
LASSO
Elastic Net
Cumulative Squared-Error Gain
700
600
500
400
300
200
100
0
−100
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
Date

A.15 Forecasting U.S. Corporate Credit Spreads Using Macroeconomic and Financial Risk Indicators





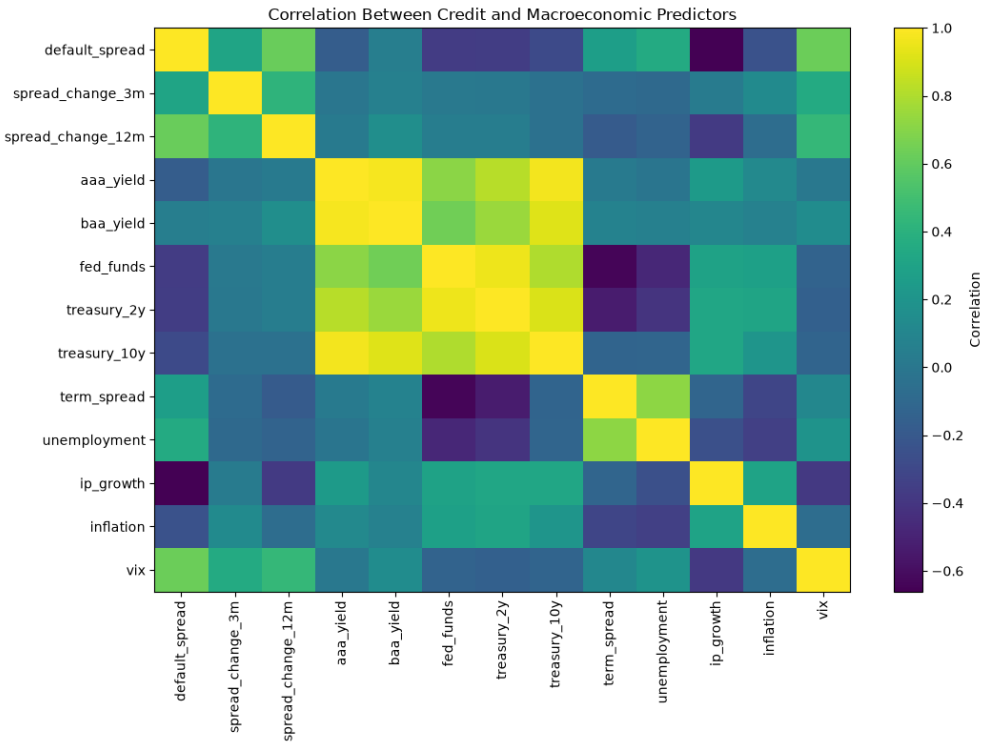


Figure 4: *Correlation Matrix for US Corporate Credit Spreads*

default_spread	-0.2912
spread_change_3m	0.0369
spread_change_12m	0.0521
aaa_yield	-0.0470
baa_yield	-0.1260
fed_funds	0.0456
treasury_2y	-0.0135
treasury_10y	0.0324
term_spread	0.1007
unemployment	-0.0292
ip_growth	-0.0934
inflation	0.1808
vix	0.0970

Figure 5: *Ridge Coefficients*

default_spread	-0.0001
spread_change_3m	0.0000
spread_change_12m	-0.0000
aaa_yield	0.0000
baa_yield	-0.0000
fed_funds	0.0000
treasury_2y	0.0000
treasury_10y	0.0000
term_spread	-0.0000
unemployment	-0.0000
ip_growth	0.0000
inflation	0.0001
vix	-0.0000

Figure 6: *OLS Coefficients*

default_spread	0.0
spread_change_3m	0.0
spread_change_12m	0.0
aaa_yield	0.0
baa_yield	0.0
fed_funds	0.0
treasury_2y	0.0
treasury_10y	0.0
term_spread	0.0
unemployment	0.0
ip_growth	0.0
inflation	0.0
vix	0.0

Figure 7: *LASSO Coefficients*

default_spread	0.0
spread_change_3m	0.0
spread_change_12m	0.0
aaa_yield	0.0
baa_yield	0.0
fed_funds	0.0
treasury_2y	0.0
treasury_10y	0.0
term_spread	0.0
unemployment	0.0
ip_growth	0.0
inflation	0.0
vix	0.0

Figure 8: *ElasticNet Coefficients*

	OLS	Ridge	LASSO	Elastic_Net	LASSO_Selected	ElasticNet_Selected
default_spread	-0.2912	-0.0001	0.0	0.0	False	False
spread_change_3m	0.0369	0.0000	0.0	0.0	False	False
spread_change_12m	0.0521	-0.0000	0.0	0.0	False	False
aaa_yield	-0.0470	0.0000	0.0	0.0	False	False
baa_yield	-0.1260	-0.0000	0.0	0.0	False	False
fed_funds	0.0456	0.0000	0.0	0.0	False	False
treasury_2y	-0.0135	0.0000	0.0	0.0	False	False
treasury_10y	0.0324	0.0000	0.0	0.0	False	False
term_spread	0.1007	-0.0000	0.0	0.0	False	False
unemployment	-0.0292	-0.0000	0.0	0.0	False	False
ip_growth	-0.0934	0.0000	0.0	0.0	False	False
inflation	0.1808	0.0001	0.0	0.0	False	False
vix	0.0970	-0.0000	0.0	0.0	False	False

Figure 9: *Coefficient Table for US Corporate Credit Spreads*

	Model	RMSE	MAE	R2
0	Historical Mean	0.2350	0.1760	-0.0240
1	OLS	0.5283	0.4646	-4.1741
2	Ridge	0.2350	0.1760	-0.0232
3	LASSO	0.2350	0.1760	-0.0240
4	Elastic Net	0.2350	0.1760	-0.0240

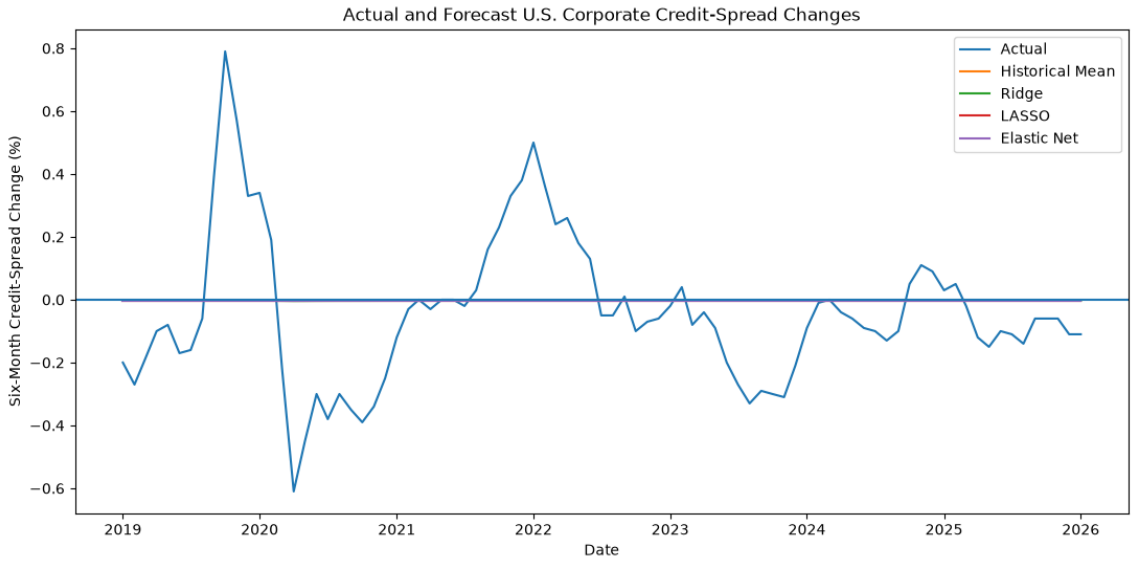
Figure 10: *Performance of Different Models*

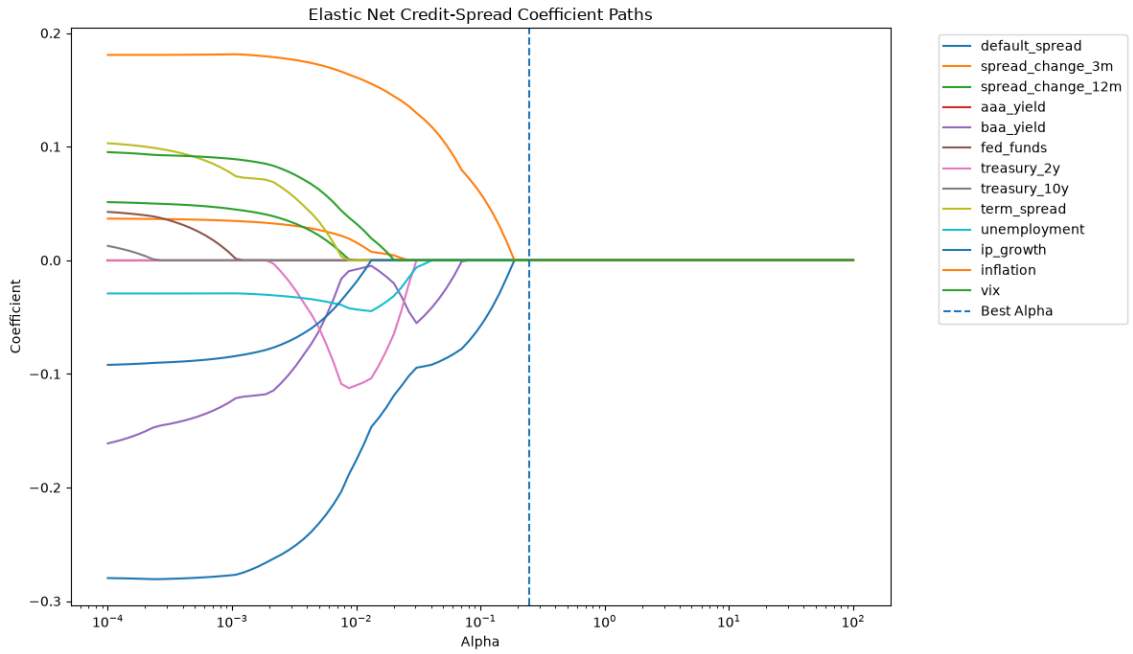
Model	OOS R^2 Value
OLS	-4.0529
Ridge	0.0008
LASSO	0.0
Elastic Net	0.0

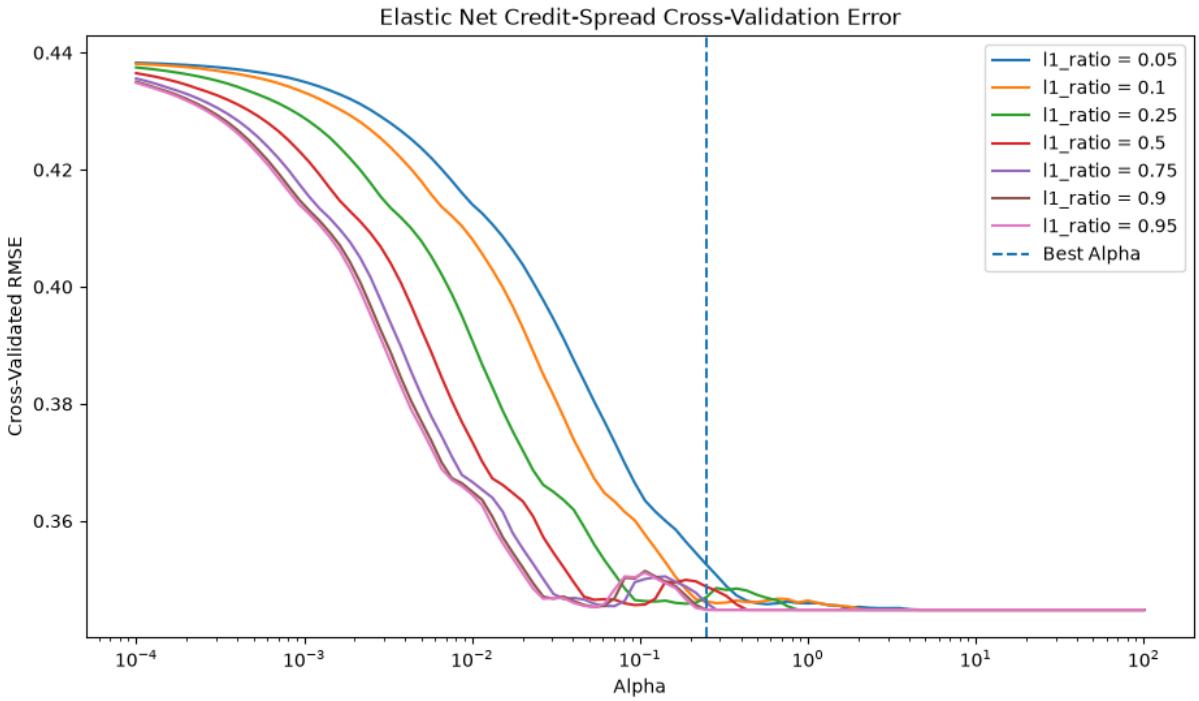
Table 3: *OOS R^2 Values for different Models*

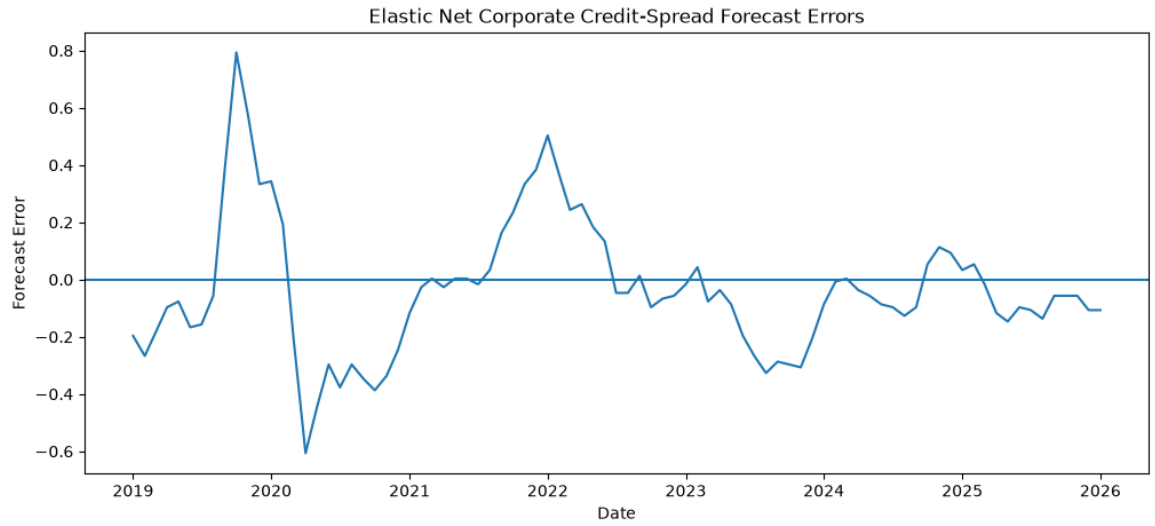
Model	Direction Accuracy
Historical Mean	67.86 %
OLS	35.71%
Ridge	67.86 %
LASSO	67.86 %
Elastic Net	67.86 %

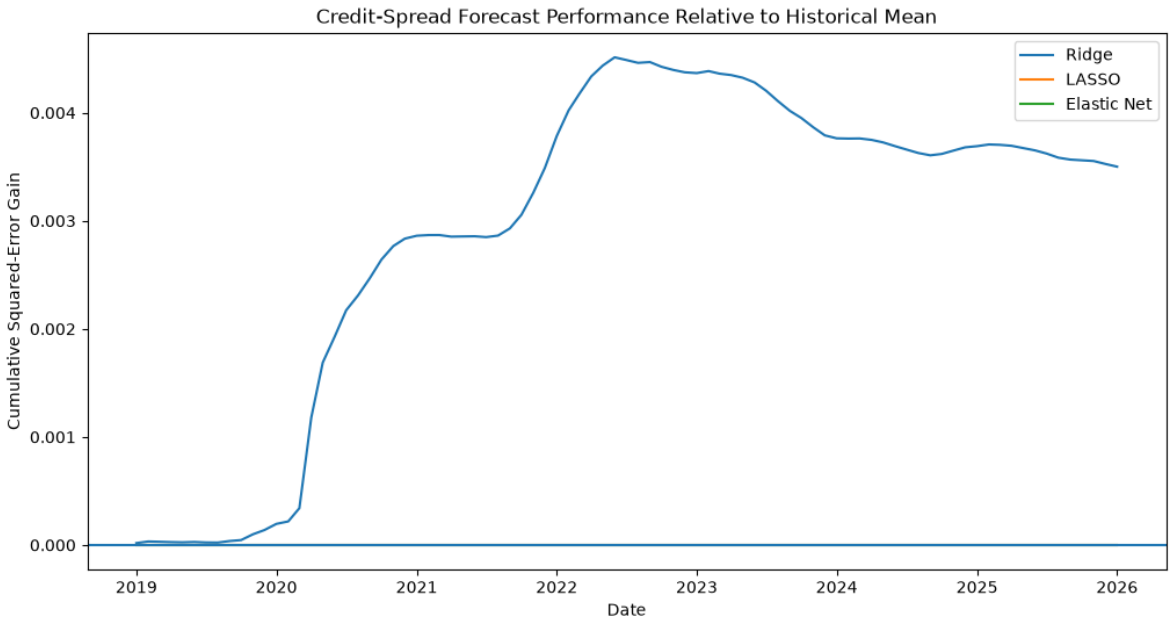
Table 4: *Direction Accuracy of Different Models*



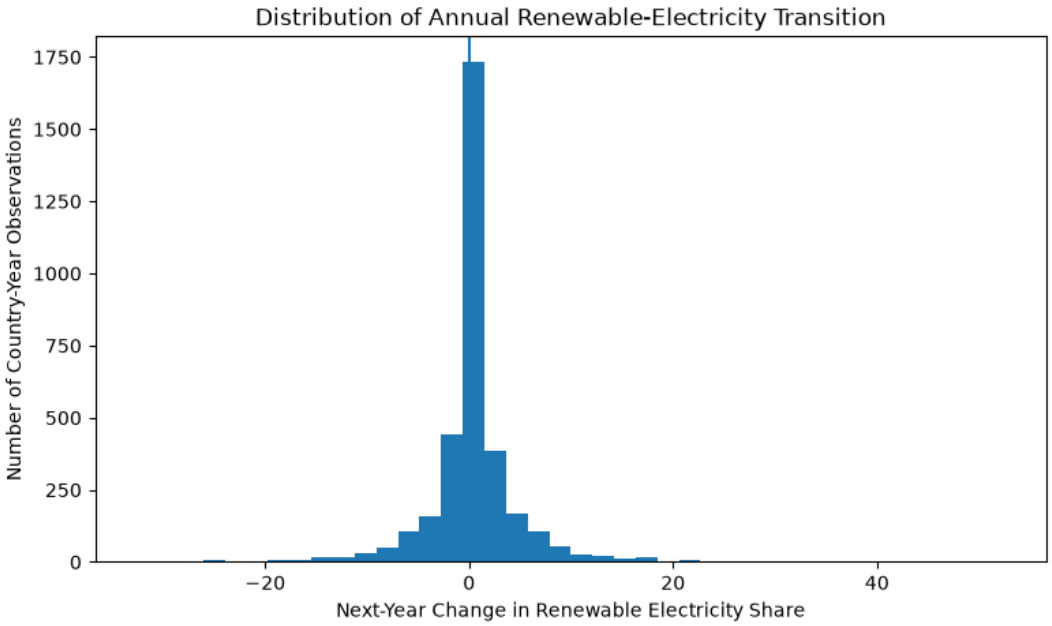

Elastic Net Credit-Spread Coefficient Paths
Coefficient
Alpha
default_spread
spread_change_3m
spread_change_12m
aaa_yield
baa_yield
fed_funds
treasury_2y
treasury_10y
term_spread
unemployment
ip_growth
inflation
vix
Best Alpha

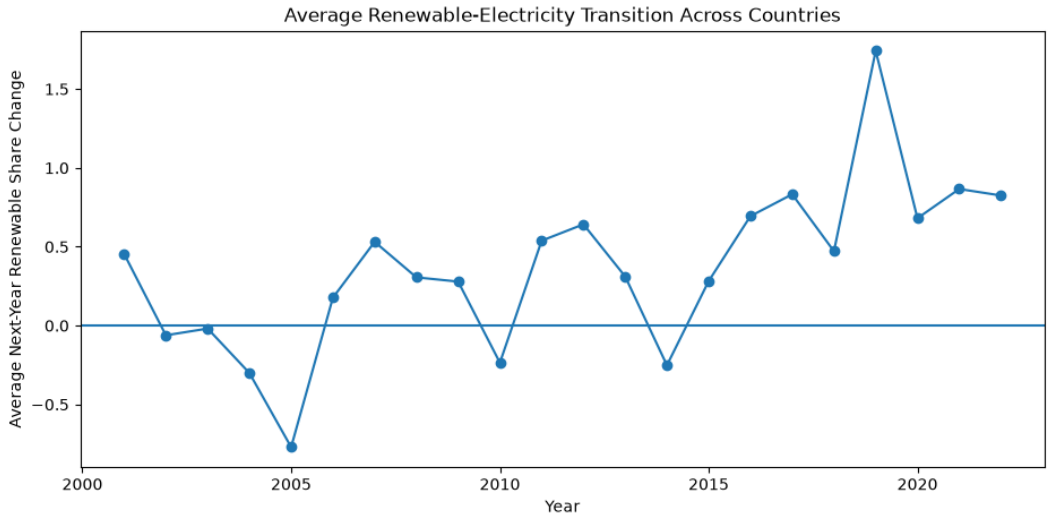

Elastic Net Credit-Spread Cross-Validation Error
Cross-Validated RMSE
Alpha
l1_ratio = 0.05
l1_ratio = 0.1
l1_ratio = 0.25
l1_ratio = 0.5
l1_ratio = 0.75
l1_ratio = 0.9
l1_ratio = 0.95
Best Alpha


Elastic Net Corporate Credit-Spread Forecast Errors
Forecast Error
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
2019
2020
2021
2022
2023
2024
2025
2026
Date


Credit-Spread Forecast Performance Relative to Historical Mean
Cumulative Squared-Error Gain
0.004
0.003
0.002
0.001
0.000
Ridge
LASSO
Elastic Net
2019
2020
2021
2022
2023
2024
2025
2026
Date

A.16 Forecasting the Renewable-Energy Transition Across Countries Using Economic, Emissions and Energy-Mix Indicators





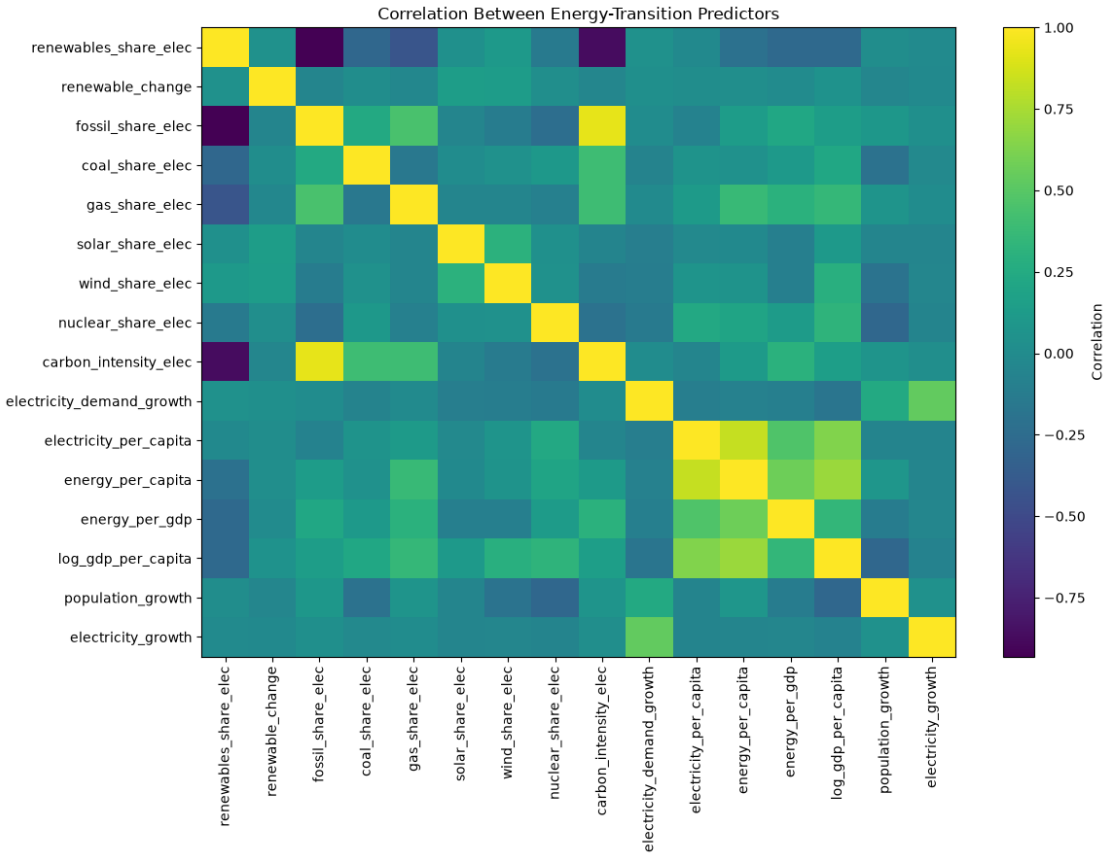


Figure 11: *Correlation Matrix for Renewable Energy Transition*

renewables_share_elec	5759.8453
renewable_change	-0.5587
fossil_share_elec	5846.2703
coal_share_elec	-0.0838
gas_share_elec	-0.1973
solar_share_elec	0.1999
wind_share_elec	0.4725
nuclear_share_elec	2194.1412
carbon_intensity_elec	-0.0017
electricity_demand_growth	-0.1964
electricity_per_capita	0.0500
energy_per_capita	0.2015
energy_per_gdp	-0.1167
log_gdp_per_capita	-0.1419
population_growth	-0.1217
electricity_growth	-0.2416

Figure 12: *OLS Coefficients*

renewables_share_elec	-0.1648
renewable_change	-0.3173
fossil_share_elec	0.1391
coal_share_elec	-0.0084
gas_share_elec	-0.0718
solar_share_elec	0.1476
wind_share_elec	0.2633
nuclear_share_elec	0.0619
carbon_intensity_elec	0.0805
electricity_demand_growth	-0.1717
electricity_per_capita	0.0315
energy_per_capita	0.0346
energy_per_gdp	-0.0441
log_gdp_per_capita	0.0212
population_growth	-0.0523
electricity_growth	-0.1609

Figure 13: *Ridge Coefficients*

renewables_share_elec	-0.3145
renewable_change	-0.3961
fossil_share_elec	0.0000
coal_share_elec	0.0000
gas_share_elec	-0.0000
solar_share_elec	0.0963
wind_share_elec	0.3556
nuclear_share_elec	0.0000
carbon_intensity_elec	0.0000
electricity_demand_growth	-0.1652
electricity_per_capita	0.0000
energy_per_capita	0.0000
energy_per_gdp	-0.0000
log_gdp_per_capita	0.0000
population_growth	-0.0000
electricity_growth	-0.1336

Figure 14: *LASSO Coefficients*

renewables_share_elec	-0.1548
renewable_change	-0.2867
fossil_share_elec	0.1154
coal_share_elec	0.0000
gas_share_elec	-0.0357
solar_share_elec	0.1297
wind_share_elec	0.2448
nuclear_share_elec	0.0442
carbon_intensity_elec	0.0557
electricity_demand_growth	-0.1604
electricity_per_capita	0.0125
energy_per_capita	0.0069
energy_per_gdp	-0.0014
log_gdp_per_capita	0.0175
population_growth	-0.0305
electricity_growth	-0.1419

Figure 15: *Elastic Net Coefficients*

	OLS	Ridge	LASSO	Elastic_Net	LASSO_Selected	ElasticNet_Selected
renewables_share_elec	5759.8453	-0.1648	-0.3145	-0.1548	True	True
renewable_change	-0.5587	-0.3173	-0.3961	-0.2867	True	True
fossil_share_elec	5846.2703	0.1391	0.0000	0.1154	False	True
coal_share_elec	-0.0838	-0.0084	0.0000	0.0000	False	False
gas_share_elec	-0.1973	-0.0718	-0.0000	-0.0357	False	True
solar_share_elec	0.1999	0.1476	0.0963	0.1297	True	True
wind_share_elec	0.4725	0.2633	0.3556	0.2448	True	True
nuclear_share_elec	2194.1412	0.0619	0.0000	0.0442	False	True
carbon_intensity_elec	-0.0017	0.0805	0.0000	0.0557	False	True
electricity_demand_growth	-0.1964	-0.1717	-0.1652	-0.1604	True	True
electricity_per_capita	0.0500	0.0315	0.0000	0.0125	False	True
energy_per_capita	0.2015	0.0346	0.0000	0.0069	False	True
energy_per_gdp	-0.1167	-0.0441	-0.0000	-0.0014	False	True
log_gdp_per_capita	-0.1419	0.0212	0.0000	0.0175	False	True
population_growth	-0.1217	-0.0523	-0.0000	-0.0305	False	True
electricity_growth	-0.2416	-0.1609	-0.1336	-0.1419	True	True

Figure 16: *Coefficient Table for Renewable Energy Transition*

	Model	RMSE	MAE	R2
0	Historical Mean	4.7964	2.6999	-0.0255
1	OLS	4.7517	2.7388	-0.0065
2	Ridge	4.6890	2.6481	0.0199
3	LASSO	4.6826	2.6329	0.0226
4	Elastic Net	4.6843	2.6344	0.0219

Figure 17: *Performance of Different Models*

Model	OOS R^2 Value
OLS	0.0186
Ridge	0.0443
LASSO	0.0469
Elastic Net	0.0462

Table 5: *OOS R^2 Values for different Models*

Model	Direction Accuracy
Historical Mean	56.02 %
OLS	56.27 %
Ridge	56.27 %
LASSO	56.40 %
Elastic Net	56.27 %

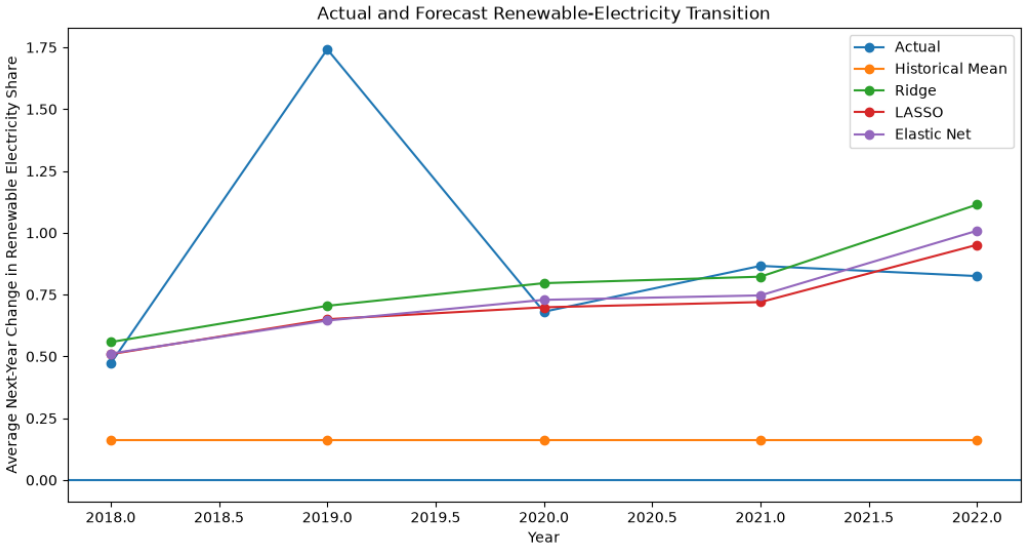
Table 6: *Direction Accuracy of Different Models*

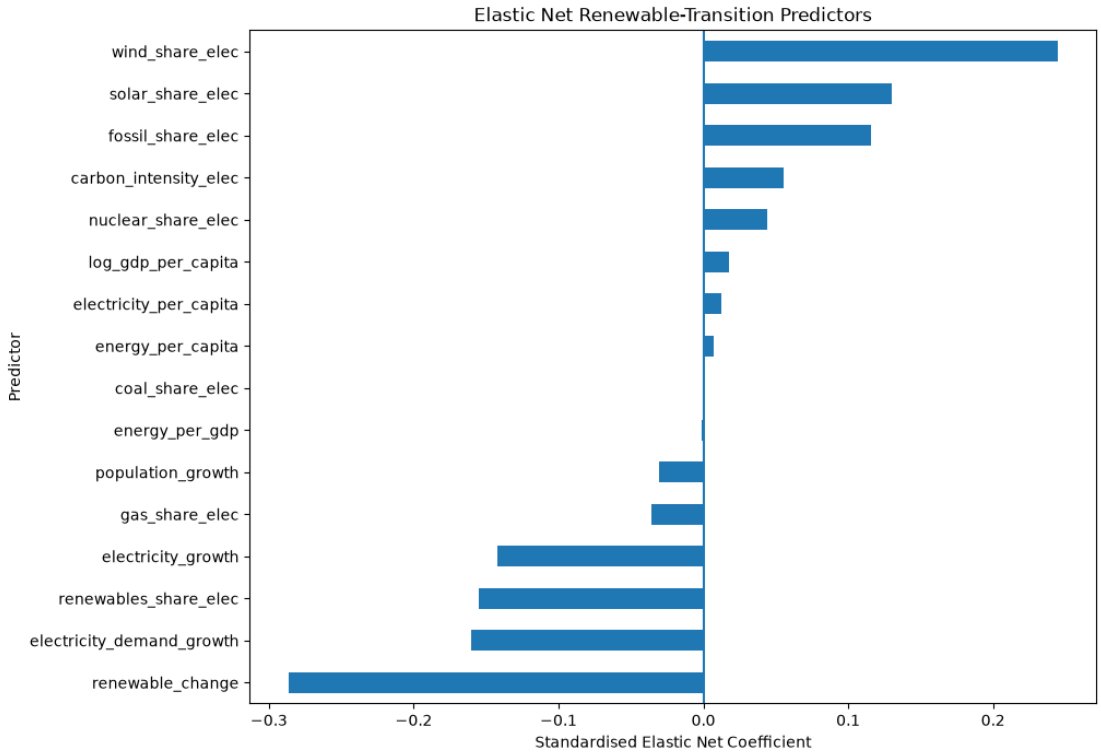
Model	RMSE	MAE	R^2
Historical Mean	4.7964	2.6999	-0.0255
OLS	4.7517	2.7388	-0.0065
Ridge	4.6890	2.6481	0.0199
LASSO	4.6826	2.6329	0.0226
Elastic Net	4.6843	2.6344	0.0219

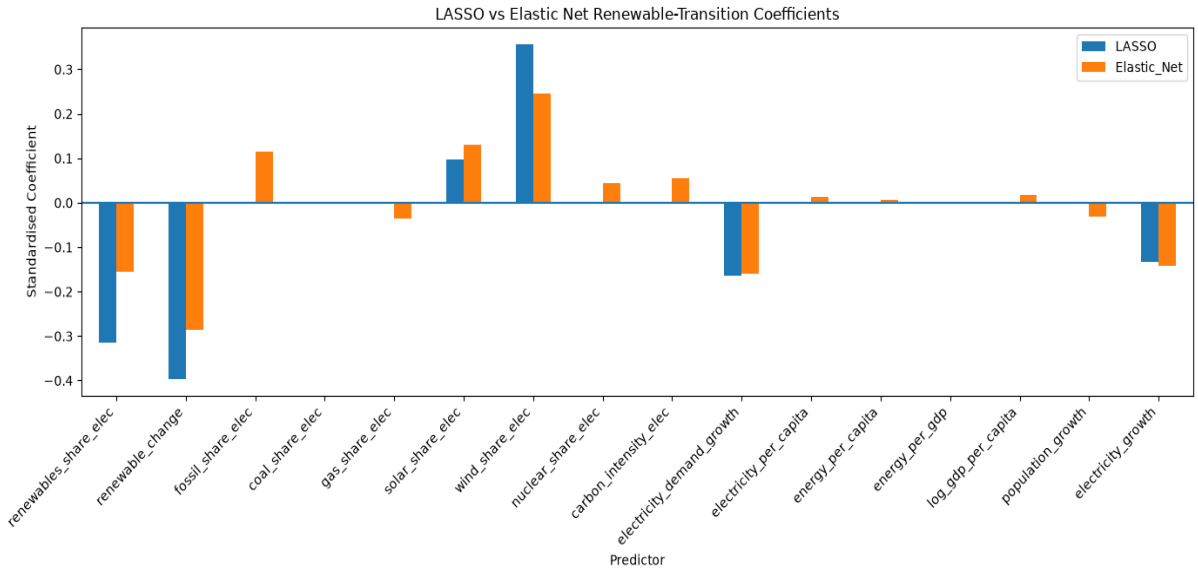
Table 7: *Forecast Results for Renewable Energy Transition*

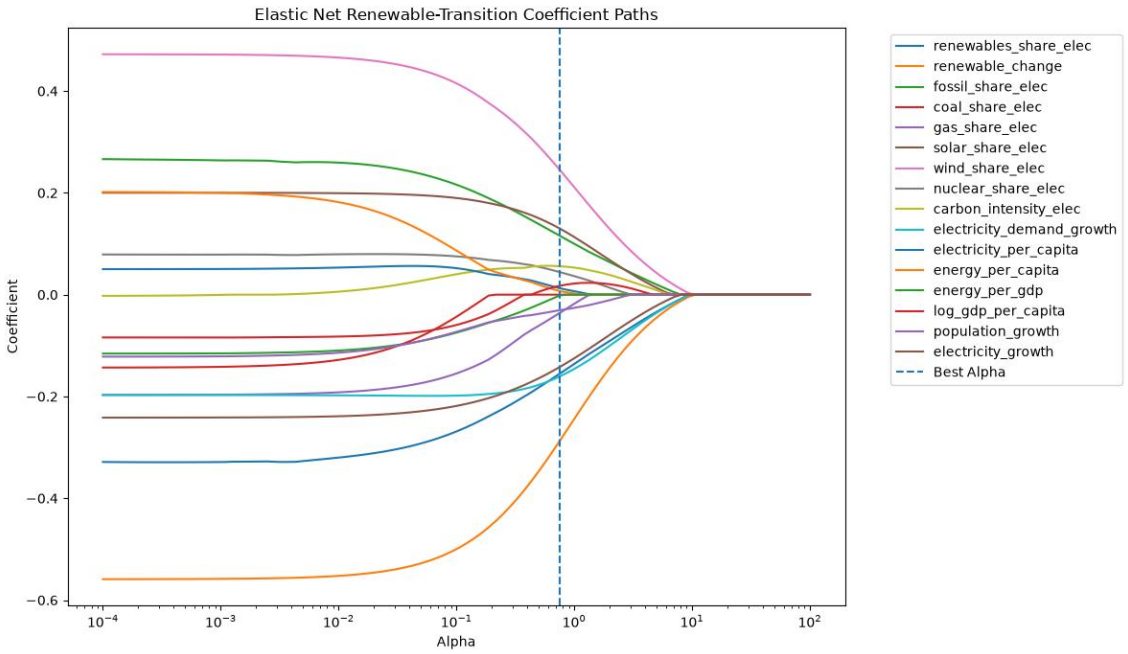
year	Actual	Historical_Mean	OLS	Ridge	LASSO	Elastic_Net
2018	0.471	0.161	0.738	0.558	0.509	0.512
2019	1.742	0.161	0.953	0.705	0.650	0.645
2020	0.681	0.161	1.051	0.796	0.699	0.729
2021	0.866	0.161	1.130	0.822	0.720	0.747
2022	0.825	0.161	1.517	1.114	0.951	1.008

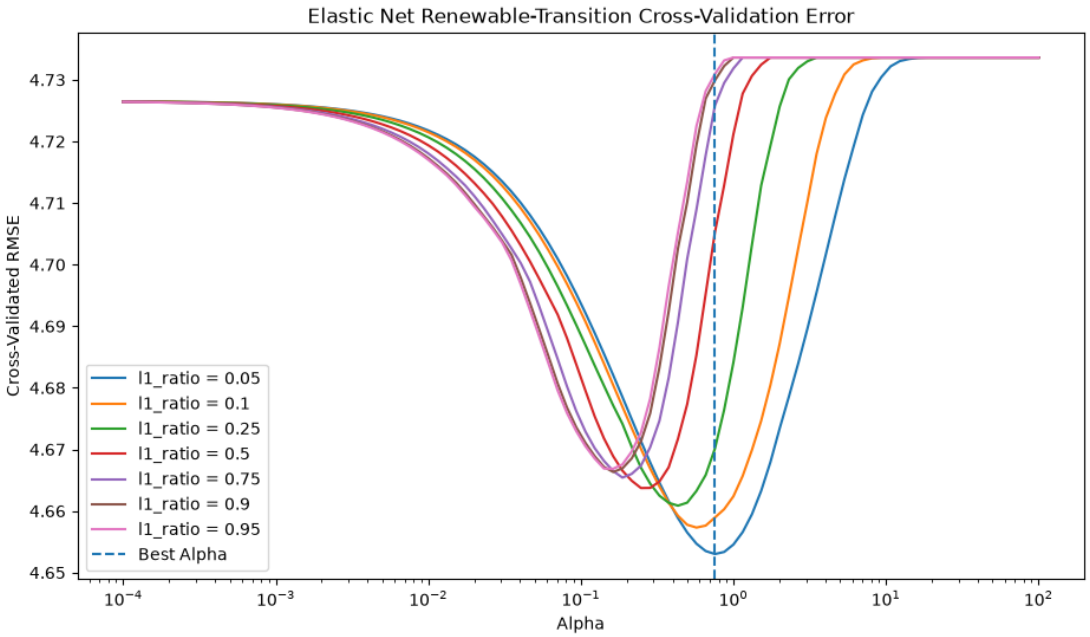
Figure 18: *Yearly Results Comparison for Renewable Energy Transition Across Countries*

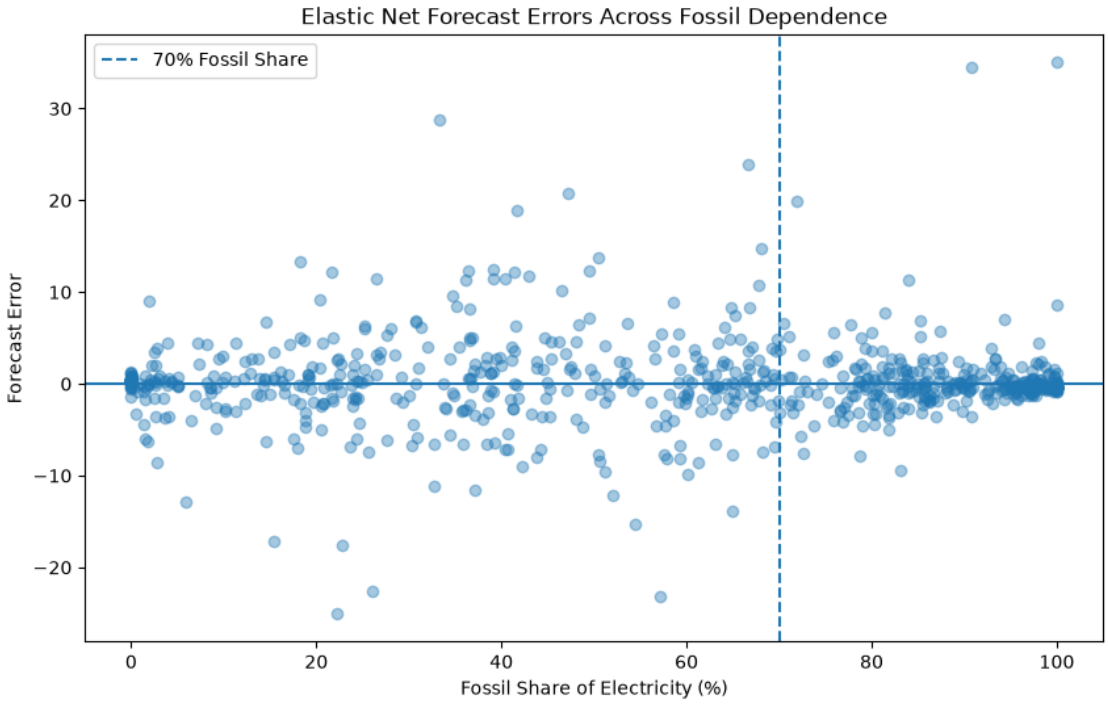


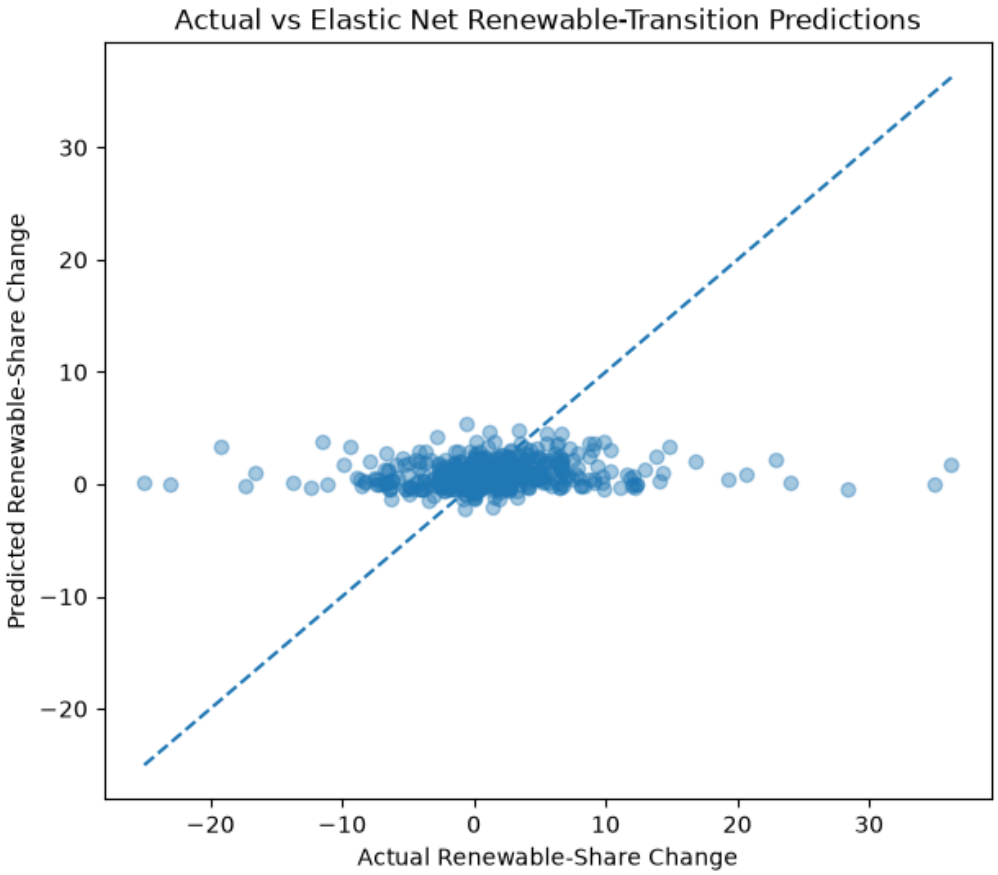

Elastic Net Renewable-Transition Predictors
wind_share_elec
solar_share_elec
fossil_share_elec
carbon_intensity_elec
nuclear_share_elec
log_gdp_per_capita
electricity_per_capita
energy_per_capita
coal_share_elec
energy_per_gdp
population_growth
gas_share_elec
electricity_growth
renewables_share_elec
electricity_demand_growth
renewable_change
Predictor
−0.3
−0.2
−0.1
0.0
0.1
0.2
Standardised Elastic Net Coefficient

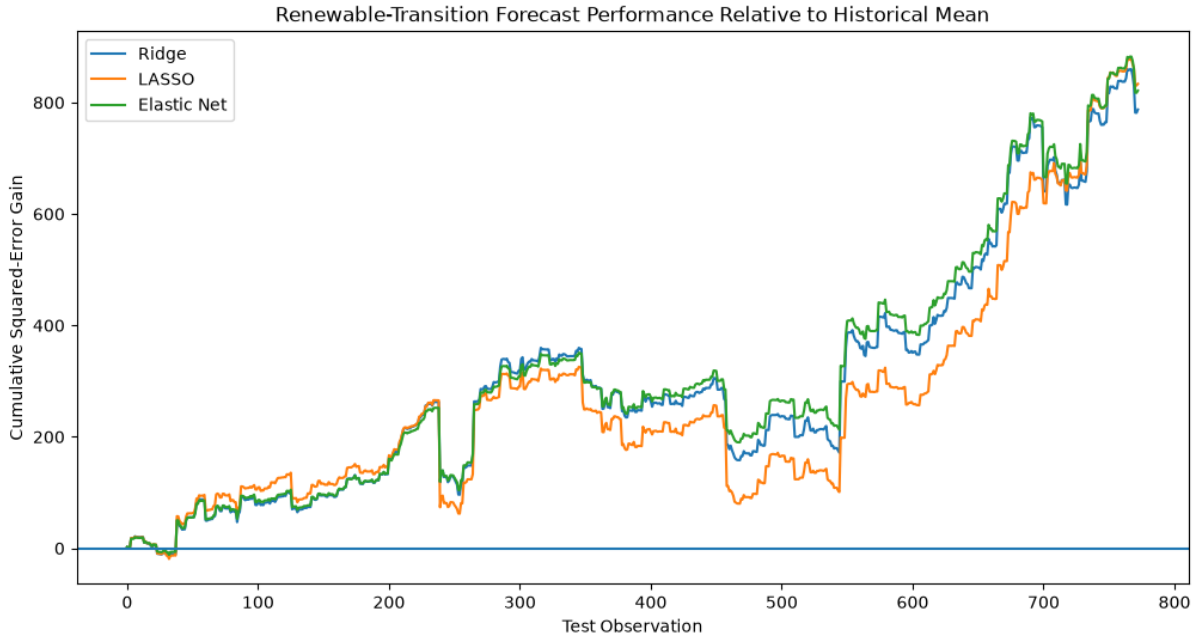

LASSO vs Elastic Net Renewable-Transition Coefficients
LASSO
Elastic_Net
Standardised Coefficient
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
renewables_share_elec
renewable_change
fossil_share_elec
coal_share_elec
gas_share_elec
solar_share_elec
wind_share_elec
nuclear_share_elec
carbon_intensity_elec
electricity_demand_growth
electricity_per_capita
energy_per_capita
energy_per_gdp
log_gdp_per_capita
population_growth
electricity_growth
Predictor


Elastic Net Renewable-Transition Coefficient Paths
Coefficient
Alpha
renewables_share_elec
renewable_change
fossil_share_elec
coal_share_elec
gas_share_elec
solar_share_elec
wind_share_elec
nuclear_share_elec
carbon_intensity_elec
electricity_demand_growth
electricity_per_capita
energy_per_capita
energy_per_gdp
log_gdp_per_capita
population_growth
electricity_growth
Best Alpha


Elastic Net Renewable-Transition Cross-Validation Error
Cross-Validated RMSE
Alpha
l1_ratio = 0.05
l1_ratio = 0.1
l1_ratio = 0.25
l1_ratio = 0.5
l1_ratio = 0.75
l1_ratio = 0.9
l1_ratio = 0.95
Best Alpha


Elastic Net Forecast Errors Across Fossil Dependence
70% Fossil Share
Forecast Error
30
20
10
0
−10
−20
0
20
40
60
80
100
Fossil Share of Electricity (%)


Actual vs Elastic Net Renewable-Transition Predictions
Predicted Renewable-Share Change
30
20
10
0
−10
−20
−20
−10
0
10
20
30
Actual Renewable-Share Change


Renewable-Transition Forecast Performance Relative to Historical Mean
Ridge
LASSO
Elastic Net
Cumulative Squared-Error Gain
0
200
400
600
800
0
100
200
300
400
500
600
700
800
Test Observation